

# DANCE: An open-source analysis pipeline and low-cost hardware to quantify aggression and courtship in *Drosophila*

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## eLife Assessment

This study presents a **valuable** open-source and cost-effective method for automating the quantification of male aggression and courtship in *Drosophila melanogaster*. The work as presented provides **solid** evidence that the use of the behavioral setup that the authors designed - using readily available laboratory equipment and standardised high-performing classifiers they developed using existing software packages - accurately and reliably characterises social behavior in *Drosophila*. The work will be of interest to *Drosophila* neurobiologists and particularly to those working on male social behaviors.

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## Abstract

Quantifying animal behaviors is pivotal for identifying the underlying neuronal and genetic mechanisms. Computational approaches have enabled automated analysis of complex behaviors such as aggression and courtship in *Drosophila*. However, existing approaches rely on rigid, rule-based algorithms and expensive hardware, limiting sensitivity to behavioral variations and accessibility. Here, we describe the DANCE (*Drosophila* Aggression and Courtship Evaluator), a low-cost, open-source platform combining machine learning-based classifiers and inexpensive hardware to quantify aggression and courtship. DANCE consists of six novel behavioral classifiers trained using a supervised machine learning algorithm. DANCE classifiers outperform existing rule-based algorithms by capturing dynamic behavioral variations. DANCE hardware is constructed using repurposed medicine blister packs and acrylic sheets, with recordings performed using smartphones, making it affordable and accessible. Benchmarking demonstrated that DANCE hardware performs comparably to sophisticated, high-cost setups. We validated DANCE in diverse contexts, including social isolation versus enrichment that modulate aggression and courtship, RNAi-mediated down-regulation of neuropeptide Dsk, and optogenetic silencing of dopaminergic neurons which

promoted aggression. DANCE provides a cost-effective and portable solution for studying *Drosophila* behaviors in resource-limited settings or closer to natural habitats. Its accessibility and robust performance democratizes behavioral neuroscience, enabling rapid screening of genes and neuronal circuits underlying complex social behaviors.

## Introduction

Detailed and accurate annotation and analysis of complex behaviors is necessary for understanding the underlying neural and molecular mechanisms. Fruit fly *Drosophila melanogaster* is one of the most accessible and well-studied model organisms for identifying neuronal and molecular underpinnings of behavior. Multiple large-scale screens have been conducted using *Drosophila*, to study complex social behaviors such as aggression and courtship (Asahina, 2017 [↗](#); Greenspan and Ferveur, 2000 [↗](#); Hall, 2002 [↗](#); Kravitz and Fernandez, 2015 [↗](#)) to identify underlying neural circuitry (Agrawal et al., 2020 [↗](#); Asahina et al., 2014 [↗](#); Davis et al., 2018 [↗](#); Hoopfer et al., 2015 [↗](#); Yadav et al., 2024 [↗](#)) and genes involved (Agrawal et al., 2020 [↗](#); Benzer, 1967 [↗](#); Gill, 1963 [↗](#); Hall, 1978 [↗](#); Ishii et al., 2022 [↗](#); Wang et al., 2008 [↗](#)). These behaviors exhibit stereotyped patterns. For instance, aggression involves chasing, fencing (Jacobs, 1960 [↗](#)), wing threats, boxing (Dow and Schilcher, 1975 [↗](#)), lunging, and tussling (Hoffmann, 1987a [↗](#), 1987b [↗](#)). Similarly, courtship consists of multiple stereotyped behaviors exhibited by the male fly, such as orienting, circling, and following the female (Cook and Cook, 1975 [↗](#); Markow, 1987 [↗](#); O'Dell, 2003 [↗](#)). To stimulate the female to be more receptive, the male produces a species-specific song by vibrating and extending its wing (Bennet-Clark and Ewing, 1969 [↗](#); Swain and von Philipsborn, 2021 [↗](#)). The male then attempts copulation by curling its abdomen and finally mounts the female for copulation (Bastock and Manning, 1955 [↗](#); Spieth, 1974 [↗](#)).

Manual analysis by trained observers is considered the gold standard in behavioral analysis, but it is a time-consuming process unsuitable for large-scale screens (Gomez- Marin et al., 2014; Robie et al., 2017a [↗](#)). ‘Computational ethology’ (Anderson and Perona, 2014 [↗](#); Datta et al., 2019 [↗](#)) helps address this challenge by automating behavioral annotation by leveraging advances in computer vision and machine learning (Robie et al., 2017b [↗](#)). This enables high-throughput behavioral screening to identify responsible genes and circuits. A typical computational ethology workflow involves recording animal behaviors and tracking their positions along with body movements. This is followed by the analysis and classification of the observed behaviors from hundreds to thousands of video frames capturing behavioral instances. Several software such as Ctrax, Caltech FlyTracker, Deep Lab Cut etc. (Branson et al., 2009 [↗](#); Eyjolfsson et al., 2014 [↗](#); Mathis et al., 2018 [↗](#)) are used widely for tracking behaviors in *Drosophila*. Each comes with their strengths and weaknesses.

For instance, Ctrax (Branson et al., 2009 [↗](#)) helps track fly body position and movements but leads to frequent identity switches among tracked flies, which is not the case while using FlyTracker (Eyjolfsson et al., 2014 [↗](#)). Effectiveness of various machine learning pipelines is eventually measured by comparing their output to human annotation, called— ‘ground- truthing’. Rule-based algorithms such as CADABRA (Dankert et al., 2009 [↗](#)) is used to quantify aggression, but it can lead to mis-scoring and identity switches, as revealed by ground-truthing (Simon and Heberlein, 2020 [↗](#)), which needs to be corrected in a semi- automated manner (Kim et al., 2018 [↗](#)). MateBook (Ribeiro et al., 2018 [↗](#)) is another rule-based algorithm to quantify courtship; however, similar to CADABRA, it tends to miss true-positive events, leading to significant mis-scoring of behaviors.

Janelia Automatic Animal Behavior Annotator (JAABA) (Kabra et al., 2013 [↗](#)) addresses the challenges of rigid rule-based approaches by employing a supervised learning approach. In the JAABA pipeline, user-labeled data is utilized for training to encompass the dynamic variations in behaviors, allowing it to predict behaviors based on learning from input data.

Recently, a few JAABA-based behavioral classifiers have been developed for measuring aggression (Chowdhury et al., 2021 [↗](#); Leng et al., 2020 [↗](#)) and courtship (Gil-Martí et al., 2023 [↗](#); Pantalia et al., 2023 [↗](#)). However, these studies either do not make these classifiers publicly available (Gil-Martí et al., 2023 [↗](#); Leng et al., 2020 [↗](#); Pantalia et al., 2023 [↗](#)) or require specialized hardware such as 3D printed parts (Chowdhury et al., 2021 [↗](#); Gil-Martí et al., 2023 [↗](#)) and expensive machine vision cameras (Chowdhury et al., 2021 [↗](#); Leng et al., 2020 [↗](#)), making them inaccessible for resource-limited settings and limiting their wider adoption.

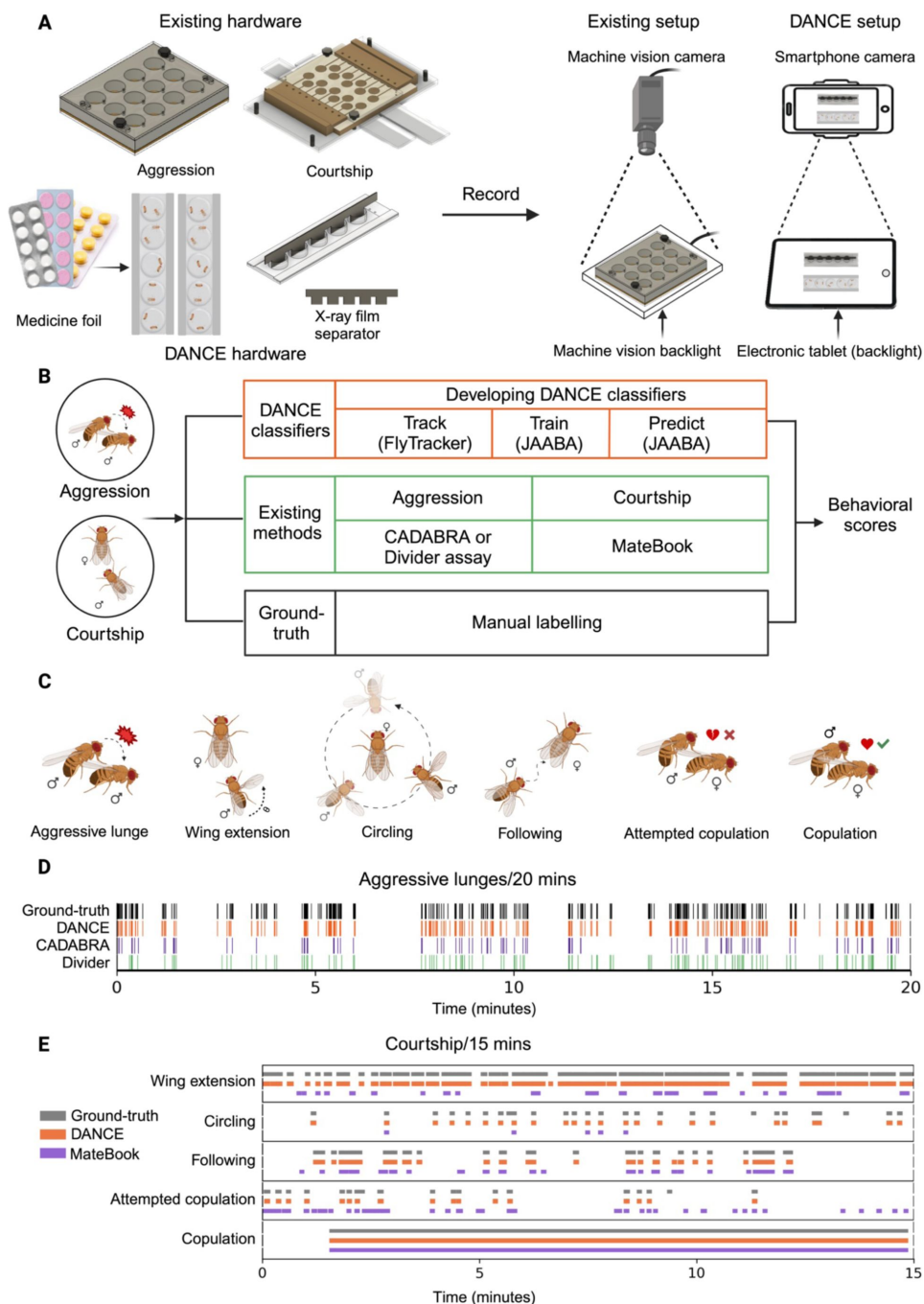
Here we describe DANCE (*Drosophila* Aggression and Courtship Evaluator), an open-source, user-friendly analysis and hardware pipeline to simplify and automate the process of robustly quantifying aggression and courtship behaviors. DANCE has two components: 1. A set of robust, machine vision-based behavioral classifiers developed using JAABA to quantify aggression and courtship; and 2. An inexpensive hardware utilizing off-the-shelf components to construct behavioral arenas and smartphone cameras to record behaviors. DANCE classifiers outperform previous methods (Dankert et al., 2009 [↗](#); Ribeiro et al., 2018 [↗](#)), improving accuracy and reliability, and DANCE hardware eliminates the need for complex, specialized behavioral arenas and expensive machine vision cameras to assess fly behaviors. Therefore, the DANCE assay can be useful for wider adoption, including in resource-strapped settings. DANCE provides a platform for rapid behavioral screening using the *Drosophila* model for discovering mechanisms underlying complex social behaviors and neurological disorders.

## Results

### DANCE assay analysis pipeline

To overcome the challenge of time-consuming manual behavioral annotation or resource-intensive, complex hardware, we developed an automated, high-throughput quantification pipeline—DANCE and trained new behavioral classifiers using an existing machine learning algorithm—JAABA (Kabra et al., 2013 [↗](#)), to robustly quantify aggression and courtship in *Drosophila* (Figure 1 [↗](#)). We built a novel hardware platform consisting of repurposed transparent medicine tablet foils/blister packs, acrylic sheets, and paper tape to easily record aggression and courtship behaviors. To record these behaviors we used Android smartphone cameras and an electronic tablet or smartphone served as a backlight illumination source (Figure 1A [↗](#), Materials and Methods). We quantify these behaviors using our ‘DANCE classifiers’. Unlike existing set-ups that cost approximately USD 3500, ‘DANCE hardware’ is fabricated using off-the-shelf components and costs less than USD 30 cents (Supplementary file 1). To benchmark the performance of the DANCE classifiers, we used pre-existing setups and rule-based methods for quantifying courtship and aggression and compared their performance with DANCE classifiers (Figure 1B-C [↗](#)).

To train DANCE classifiers using JAABA (Kabra et al., 2013 [↗](#)), for aggressive lunges, we used a pre-existing setup described in (Dankert et al., 2009 [↗](#)) modified from (Dierick, 2007 [↗](#)) (Figure 1A [↗](#); Figure 1—figure supplement 1 [↗](#)); and for courtship behaviors, we used a pre-existing set up described in (Koemans et al., 2017 [↗](#)) (Figure 1A [↗](#); Figure 1—figure supplement 2 [↗](#)). We tracked position, motion, and interactions of pairs of flies across video frames using the Caltech FlyTracker (Eyjolfsson et al., 2014 [↗](#)). To avoid data leak, we randomly divided acquired videos into two categories: “training videos” and “test videos” to train and evaluate DANCE classifiers. These test videos were also manually ‘ground-truthed’ frame by frame, which is considered the gold standard for behavioral annotation (Figure 1B, D-E [↗](#)). We benchmarked the performance of DANCE classifiers using existing rule-based algorithms, CADABRA (Dankert et al., 2009 [↗](#)) and MateBook (Ribeiro et al., 2018 [↗](#)); and an existing JAABA aggression classifier (Chowdhury et al., 2021 [↗](#)). We then compared their output against manual ground-truth. Figure 1 D-E [↗](#) show examples of courtship and aggression video annotation, suggesting that DANCE classifiers



**Figure 1.**

### DANCE assay overcomes the limitation of existing methods for quantification of aggression and courtship behaviors.

(A) Existing hardware for acquisition of aggression and courtship behavior using machine vision cameras and the simplified DANCE hardware. (B) Steps for developing DANCE classifiers and benchmarking with existing methods and manual ground-truthing to produce behavioral scores. (C) Various behavioral classifiers developed to quantify male aggression (lunge) and male courtship (wing extension, circling, following, attempted copulation, and copulation). (D) Raster plots comparing performance of ground-truth vs. DANCE classifier vs. CADABRA vs. Divider assay for aggression in a representative video. (E) Raster plots for various courtship behaviors comparing performance of ground-truth vs. DANCE classifiers vs. MateBook from representative videos. Created in *BioRender*.

outperform existing methods and are comparable with manual ground-truth. Subsequent sections describe the quantitative analysis of individual DANCE classifiers and benchmarking of DANCE hardware.

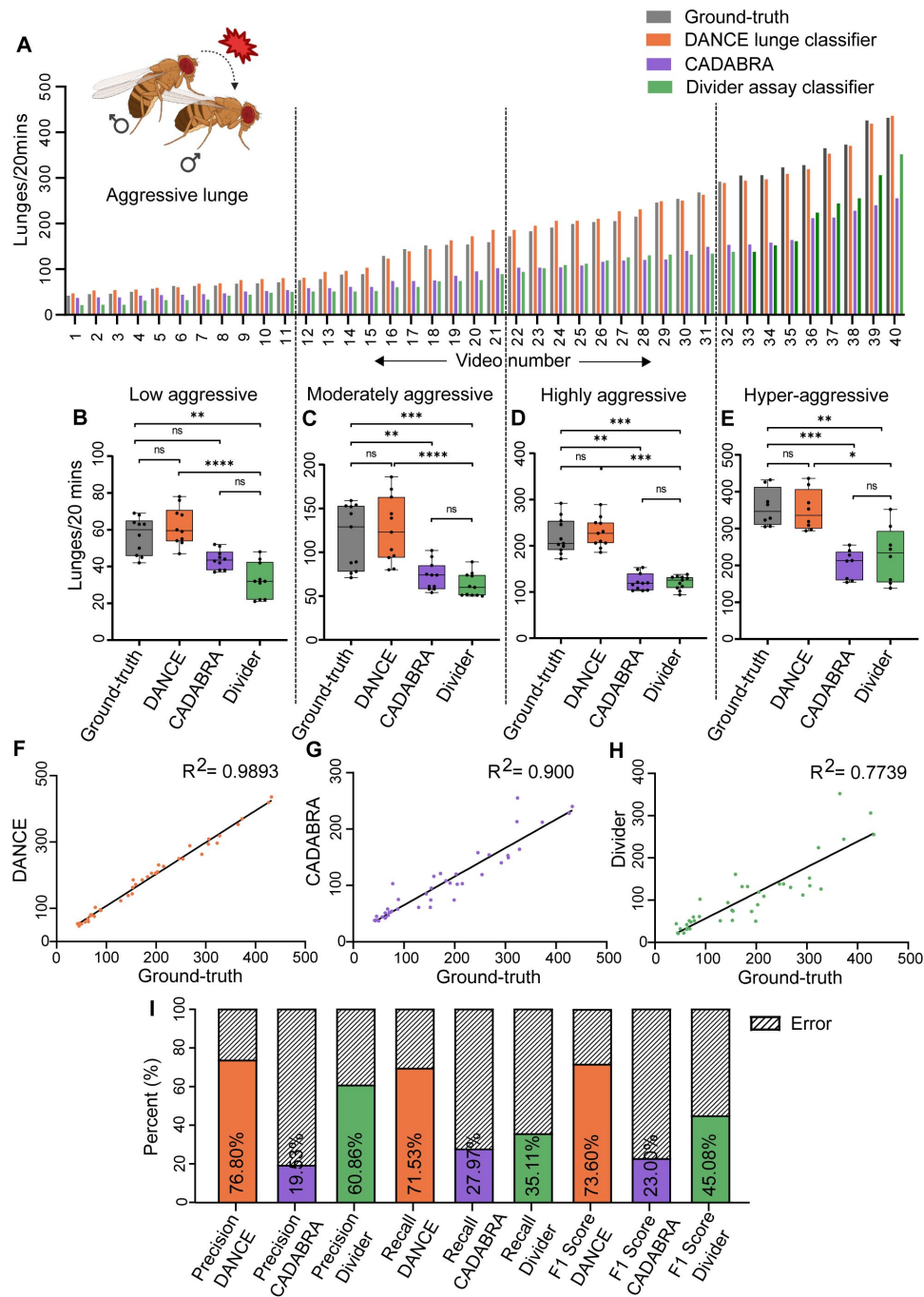
## DANCE lunge classifier to quantify aggressive behavior

Aggression is an innate, complex behavior, and *Drosophila* males exhibit several stereotyped behavioral patterns during aggressive encounters, with lunging used widely as a measure of overall aggression in males (Agrawal et al., 2020 [DOI](#); Asahina et al., 2014 [DOI](#); Chiu et al., 2021 [DOI](#); Chowdhury et al., 2021 [DOI](#); Davis et al., 2014 [DOI](#); Dierick, 2007 [DOI](#); Hoffmann, 1987a [DOI](#); Hoopfer et al., 2015 [DOI](#); Hoyer et al., 2008 [DOI](#); Jung et al., 2020 [DOI](#); Nilsen et al., 2004 [DOI](#); Watanabe et al., 2017 [DOI](#); Yadav et al., 2024 [DOI](#)). A lunge is defined as a male fly raising its front legs and hitting down on the other fly. We developed a new classifier using JAABA (Kabra et al., 2013 [DOI](#)) to robustly quantify aggressive lunges in *Drosophila*, hereafter referred to as the ‘DANCE lunge classifier.’ We quantified lunges using our classifier from 20-minute-long videos and compared the output with manual ground-truthing and existing methods— CADABRA (Dankert et al., 2009 [DOI](#)) and ‘Divider assay classifier’ (Chowdhury et al., 2021 [DOI](#)). CADABRA tends to miss several true positive lunges, likely due to its rigid, rule-based framework, which cannot adapt to the dynamic variations in behavior. To address this, ‘Divider assay classifier’ was developed based on JAABA, but the setup requires use of 3-D printed rectangular arena and dividers as well as expensive machine vision cameras (Chowdhury et al., 2021 [DOI](#)).

**Figure 2A** [DOI](#) shows lunge scores from 40 different videos using ground-truth, DANCE lunge classifier, CADABRA, and Divider assay classifier. While the ground-truth and DANCE classifier’s output are comparable, CADABRA and the Divider assay classifier underscored lunges across videos. We ground truthed the DANCE lunge classifier against 40 ‘test videos’ (**Figure 2A** [DOI](#), Materials and Methods). To address potential observer bias during ground-truthing, we compared the annotations between two separate evaluators for the DANCE lunge classifier and found no significant differences (**Figure 2— figure supplement 1A** [DOI](#)). Since aggressive lunges have a large dynamic range, we benchmarked our classifier across a range of aggressive behaviors, and subdivided the ground truthed videos into four categories: 1. low aggressive, 0–70 lunges (**Figure 2B** [DOI](#)); 2. moderately aggressive, 71–160 lunges (**Figure 2C** [DOI](#)); 3. highly aggressive, 161–300 lunges (**Figure 2D** [DOI](#)); and 4. hyper- aggressive, >300 lunges (**Figure 2E** [DOI](#)). We found that DANCE lunge classifier and manual ground-truth performance was comparable across all four categories (**Figure 2 B-E** [DOI](#)). In contrast, the CADABRA and Divider assay classifier showed significant underscoring from the manual ground-truth (**Figure 2B-E** [DOI](#)), however, a strong correlation was observed between our DANCE lunge classifier and ground-truth (**Figure 2F** [DOI](#), Supplementary file 2), whereas, CADABRA and the Divider assay classifier showed only a weaker correlation with the ground truth (**Figure 2G-H** [DOI](#), Supplementary file 2). We reasoned that CADABRA’s lower performance is most likely due to the rigid rules used to define a lunge (Dankert et al., 2009 [DOI](#)) (Materials and Methods); and the Divider assay classifier’s trained in a rectangular arena did not perform as well in a circular arena since JAABA classifiers utilize features related to arena geometry.

To further evaluate the performance of our DANCE lunge classifier, we used three metrics— precision, recall, and F1 score and compared the output of the classifier and manual ground-truth across 40 test videos. We calculated the counts of true positives, false positives, and false negatives to derive precision, which depicts the ratio of true positives among overall predictions from the classifier; recall, which depicts the ratio of true positives among the ground-truth scores; and F1 score, which is a harmonic mean of precision and recall (**Figure 2I** [DOI](#)) (Materials and Methods). Our DANCE lunge classifier had a precision of 76.80%, a recall rate of 71.53% and an overall F1 score of 73.60% (**Figure 2I** [DOI](#)), which is higher than the CADABRA and the Divider assay suggesting that the DANCE lunge classifier outperforms these existing methods. Together, our analysis suggests that the DANCE lunge classifier performs with high precision and quantifies lunge numbers robustly over a broad range of fighting intensities.





**Figure 2.**

### DANCE lunge classifier outperforms existing methods of quantifying male aggression.

(A) Lunge scores from 20 minute long videos from ground-truth (grey), DANCE lunge classifier (orange), CADABRA (purple) and Divider assay classifier (green). (B-E) Comparison of lunge scores from different levels of aggression. Male flies showing lunges ranging from 0 to 500 were compared manually, with a new DANCE lunge classifier, CADABRA, and Divider assay classifier. (B) 'low aggressive', (n=10; \*\*P<0.0017, \*\*\*\*P<0.0001). (C) 71–160 lunges, 'moderately aggressive', (n=11; \*\*P<0.0102, \*\*\*\*P<0.0002, \*\*\*\*P<0.0001). (D) 161–300 lunges, 'highly aggressive', (n = 11; \*\*P<0.0057, \*\*\*\*P<0.0002, \*\*\*\*P<0.0004). and (E) >300 lunges, 'hyper-aggressive', (n = 8; \*P<0.0402, \*\*P<0.0029, \*\*\*\*P<0.0006; Friedman's ANOVA with Dunn's test) (F) Regression analysis of DANCE 'lunge classifier' vs. manual scores ( $R^2=0.9893$ , n=40). (G) CADABRA vs. DANCE lunge classifier ( $R^2=0.9$ , n=40). (H) Divider assay lunge classifier vs. manual scores ( $R^2=0.7739$ , n=40). (I) F1 score, precision, and recall between DANCE lunge classifier, CADABRA, and Divider assay classifier.

## DANCE classifiers to quantify courtship behaviors in *Drosophila*

The first report of *Drosophila* courtship behavior, identified stereotypic behaviors such as wing ‘scissor-like’ movements, males ‘swaying around the female,’ licking, tapping, and mounting (Sturtevant, 1915 [↗](#)). By the 2000s, studies uncovered genes and neural circuits involved in courtship (Dickson, 2008 [↗](#); Pavlou and Goodwin, 2013 [↗](#)). Automated analysis techniques for courtship exist but face barriers to adoption due to expensive hardware, reliance on 3D printing, or lack of publicly available analysis code or classifiers (Gil-Martí et al., 2023 [↗](#); Koemans et al., 2017 [↗](#); Leng et al., 2020 [↗](#); Reza et al., 2013 [↗](#)).

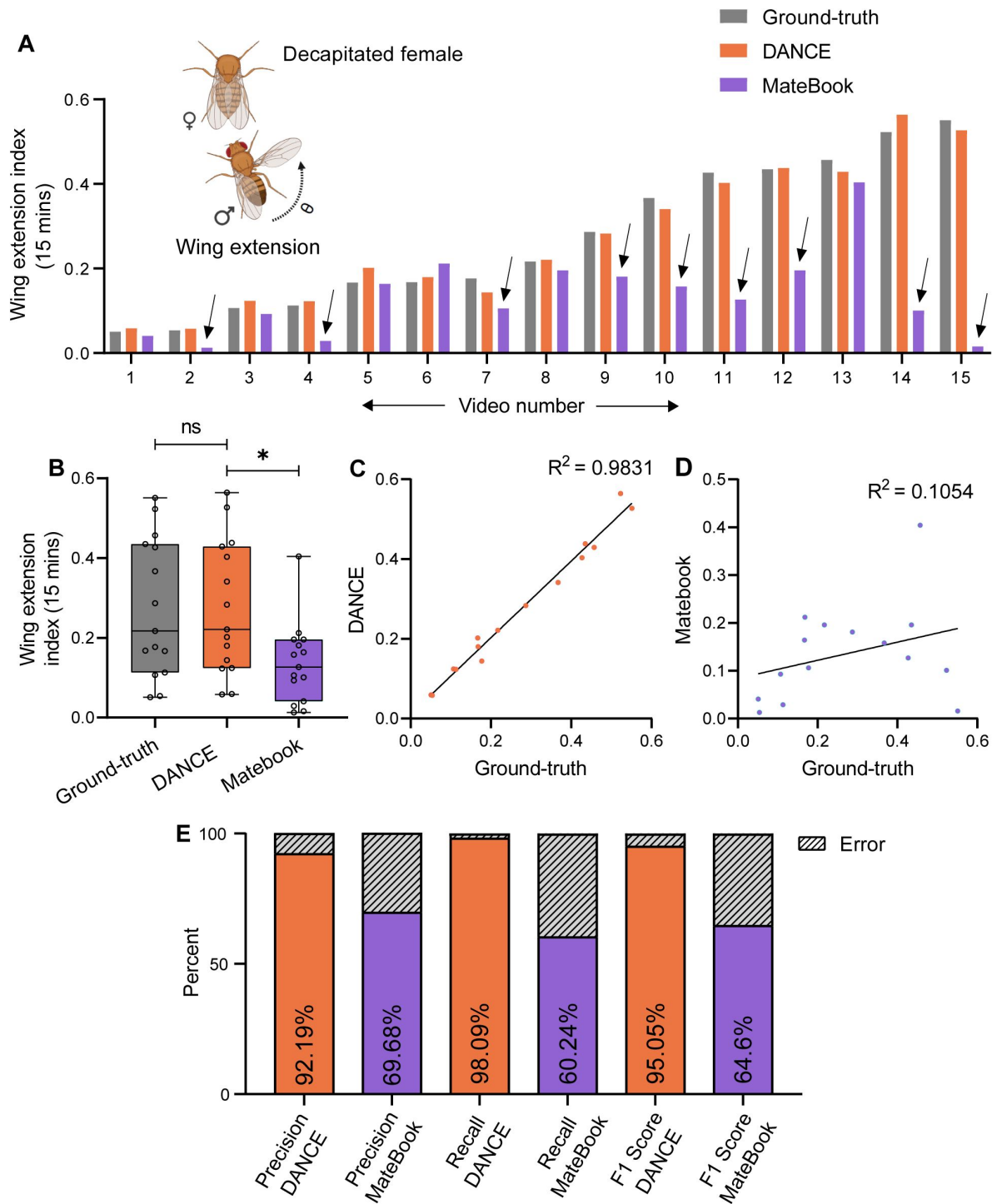
MateBook is a recent rule-based algorithm that was developed to automate quantifying various courtship behaviors in *Drosophila* males (Ribeiro et al., 2018 [↗](#)). It relies on predefined mathematical rules based on CADABRA (Dankert et al., 2009 [↗](#)). To resolve ambiguities in the two overlapping flies during estimating their trajectories from the video recordings, it assigns the identities by relying on the distinct body sizes of the male and female fly, as the females are larger than males. However, decapitated virgin females are of similar body size as males, and are often used in courtship assays to assess the male attractiveness to other factors such as pheromones, independent of the female behavior (Cook and Cook, 1975 [↗](#); Spieth, 1966 [↗](#)). In such cases, rigid rule-based approach can introduce both false positives and false negatives since the condition about differential body size is not met.

To address the inherent inflexibility of rule-based algorithms, we trained and validated five new courtship classifiers using JAABA (Kabra et al., 2013 [↗](#)) to provide a detailed and robust analysis of multiple steps of courtship ritual (Sokolowski, 2001 [↗](#)) including— wing extension, following, circling, attempted copulation, and copulation. We evaluated the performance of our DANCE classifiers by comparing the output to ground-truth and MateBook (Ribeiro et al., 2018 [↗](#)). Criteria used to develop various classifiers are described in Materials and Methods. To compare the output, we calculated a behavior index, since courtship behaviors, unlike aggressive lunges, are of varied duration. To address potential observer bias during ground-truthing, we compared the annotations between two separate evaluators for the DANCE courtship classifiers and found no significant differences, suggesting robustness of our classifiers (**Figure 2— figure supplement 1B-F** [↗](#)).

### Wing Extension

During unilateral wing extension, the male vibrates its wing at a specific frequency to produce a species-specific courtship song to attract the female (Shorey, 1962 [↗](#); Spieth, 1952 [↗](#)). **Figure 3A** [↗](#) shows wing extension index from the ground-truth (grey) and the ‘DANCE wing extension classifier’ (orange), which are comparable across 15 videos from males courting decapitated virgin females, however, MateBook’s output (purple) seems to underscore these events in most videos (**Figure 3A** [↗](#), Friedman’s ANOVA with Dunn’s test; ground-truth vs. MateBook,  $p=0.0020$ ,  $n=15$ ; ground-truth vs. DANCE classifier,  $p>0.9999$ ) (**Figure 3A** [↗](#)). We also evaluated the overall wing extension index from all 15 videos and found ground-truth scores and the DANCE wing extension classifier matched well, but MateBook significantly underscoring wing extension (**Figure 3B** [↗](#)). We found a strong correlation between the DANCE wing extension classifier index to ground-truth (**Figure 3C** [↗](#)) but a weaker correlation with MateBook (**Figure 3D** [↗](#)). To further evaluate our classifier’s performance, we calculated precision, recall, and F1 scores, as described earlier. The DANCE wing extension classifier had a precision of 92.19%, a recall rate of 98.09%, and an overall F1 score of 95.05% when compared to the ground-truth which is higher than the MateBook (**Figure 3E** [↗](#)).

Quantification of the videos containing mated females identified a robust correlation between our classifier’s output and the ground-truth, whereas a general trend of underscoring was observed for MateBook (**Figure 3— figure supplement 1** [↗](#)). Together, this suggests that the DANCE wing



**Figure 3.**

### DANCE wing extension classifier outperforms existing methods of quantification.

(A) Wing extension index of males from 15 minutes long videos from the ground-truth (grey), DANCE wing extension classifier (orange) and MateBook (purple), against decapitated virgin females. MateBook underscored wing extension across multiple videos (black arrows). (B) Comparison of ground-truth vs. DANCE vs. MateBook wing extension classifier (Kruskal-Wallis ANOVA with Dunn's test, ns,  $p > 0.9999$ ,  $*p = 0.0436$ ;  $n = 15$ ). (C) Regression analysis of the DANCE wing extension classifier vs. ground-truth ( $R^2 = 0.9831$ ,  $n = 15$ ). (D) MateBook vs. ground-truth ( $R^2 = 0.1054$ ,  $n = 15$ ). (E) F1 score, precision, and recall of DANCE wing extension classifier and MateBook against ground-truth scores.



extension classifier is robust for both decapitated and mated females and performs with high precision.

## Attempted copulation and copulation

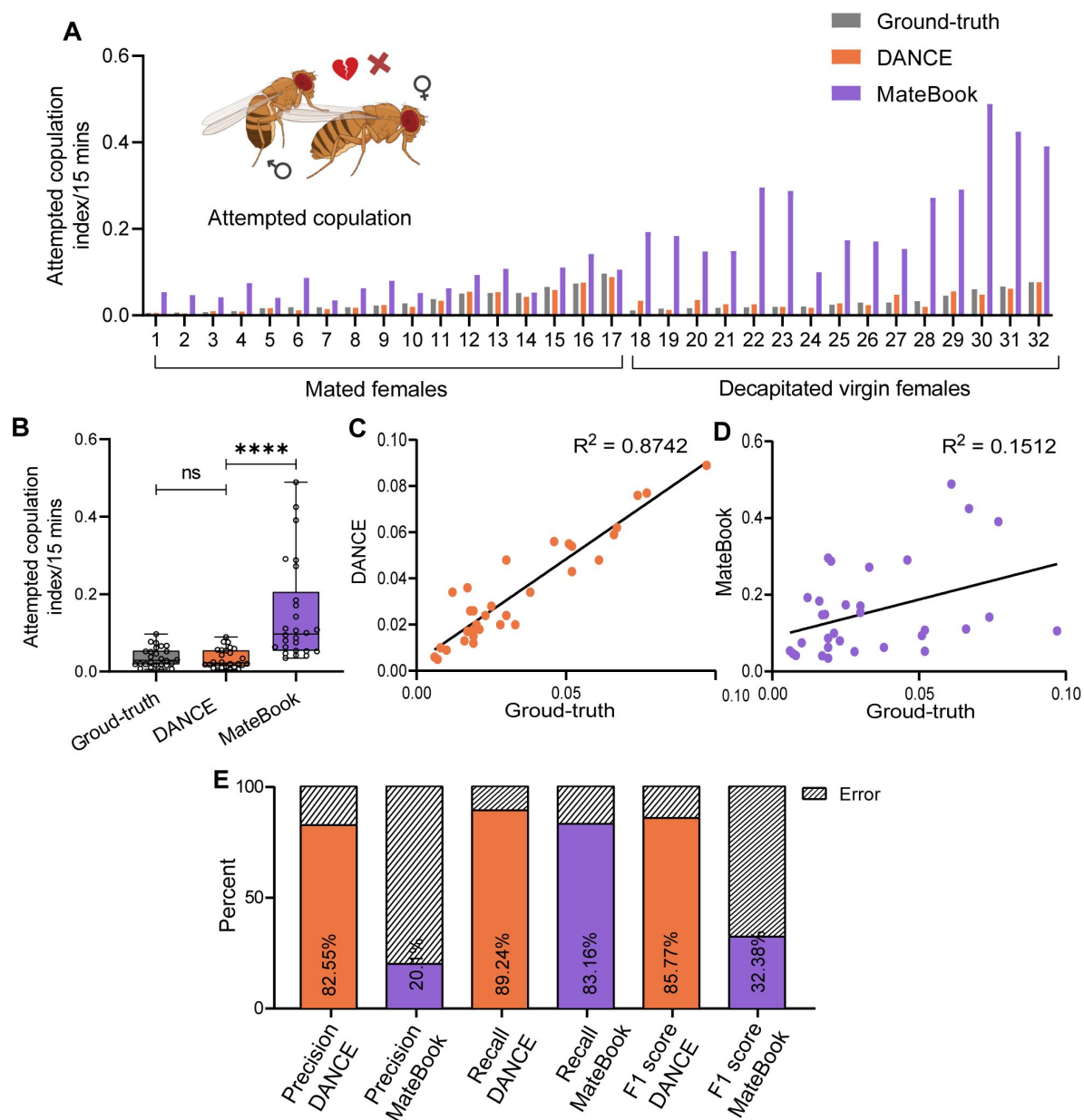
Copulation in male flies lasts for about 15–25 minutes, and its duration is primarily determined by the male (MacBean and Parsons, 1967 [↗](#)). Interrupted mating experiments show that the sperms are transferred several minutes after copulation begins (Fowler, 1973 [↗](#); Tompkins et al., 1980 [↗](#)). Thus this differentiates the mounting behavior into two— successful copulation and unsuccessful attempted-copulation. We developed the ‘DANCE attempted copulation’ with a threshold of 0.33 to <45 seconds and the ‘DANCE copulation’ classifiers with a threshold of >45 seconds, based on existing definitions (Ribeiro et al., 2018 [↗](#)), but with modifications, since MateBook only quantifies ‘copulation’ and not ‘attempted-copulation’ (Materials and Methods). For ground-truthing, we combined both the dataset of mated and decapitated virgin females, because we observed that MateBook consistently overscored in both datasets. (Figure 4A–B [↗](#)). MateBook’s rule-based approach led to significant overscoring with reduced thresholds for attempted copulations, especially with decapitated virgins, due to both increased false positives and identity switches. As expected, the DANCE classifier matched well with the ground-truth scores for each individual video (Figure 4A— Friedman’s ANOVA with Dunn’s test;  $p > 0.9999$ ,  $n = 32$ ), as well as the overall attempted copulation index (Figure 4B [↗](#)). We found a strong correlation between ground-truth and the DANCE attempted copulation classifier index (Figure 4C [↗](#)) but a weaker correlation with MateBook (Figure 4D [↗](#)). The DANCE classifier performed robustly with a precision rate of 82.55%, a recall rate of 89.24%, and a F1 score of 85.77% as compared to MateBook (Figure 4E [↗](#)). Together, this suggests that the DANCE attempted copulation outperforms existing rule based framework and is able to capture dynamic variations in behaviors robustly.

Since the male readily copulates with the virgin females for several minutes, we used videos where males were paired with virgin females to train the ‘DANCE copulation classifier’. Our classifier matched well with the ground truth in the number of bouts, with a 100% precision, recall and F1 score (Figure 4— figure supplement 1 [↗](#)). The rule-based approach of MateBook also works well with copulation, which MateBook defines as ‘occlusion’ persisting for a long time (Materials and Methods). However sometimes, misaligned videos can cause ‘arena detection’ errors, causing missed events (Figure 4— figure supplement 1A [↗](#), video#18) or false positives (Figure 4— figure supplement 1 [↗](#), video#13) leading to lower precision, recall and F1 scores.

## Circling

The male circles around the female, which is a unique male behavior during courtship. Studies have shown that a refractory female gets restimulated by the circling behavior of the males (Kessler, 1962 [↗](#)). In addition, differences in the frequency of circling behavior and other courtship rituals is used to maintain reproductive isolation among different species (Brown, 1965 [↗](#)).

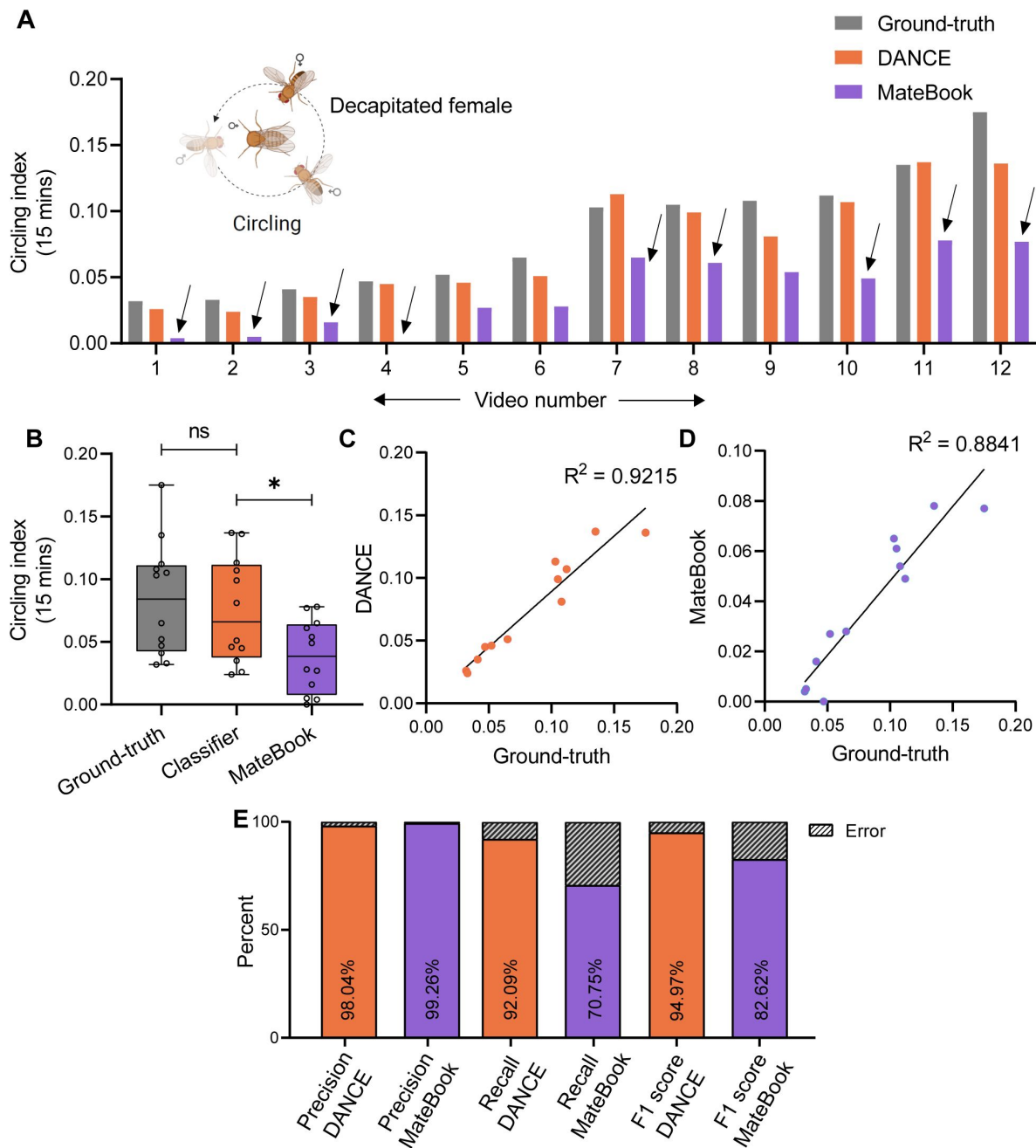
To evaluate the ‘DANCE circling classifier’, we used both the decapitated virgin female (Figure 5 [↗](#)) and mated female dataset (Figure 5— Figure Supplement 1 [↗](#)). The circling index from the ground-truth and the DANCE classifier output are comparable; however, MateBook underscoring these events (Figure 5A [↗](#)), most likely due to identity switches. DANCE circling classifier matched well from the ground-truth for individual videos (Figure 5A [↗](#), Friedman’s ANOVA with Dunn’s test;  $p < 0.2049$ ,  $n = 12$ ). Overall circling index from all 12 videos from the ground-truth scores and the DANCE circling classifier were comparable (Figure 5B [↗](#)), whereas MateBook significantly underscores circling in individual videos (Figure 5A [↗](#), Friedman’s ANOVA with Dunn’s test;  $***p < 0.0001$ ,  $n = 12$ ). We found a strong correlation between the DANCE circling classifier index to ground-truth (Figure 5C [↗](#)) but a weaker correlation with MateBook (Figure 5D [↗](#)). The DANCE circling classifier has a precision rate and a recall rate of 98.04% and 92.09%, with a F1 score of 94.97% (Figure 5E [↗](#)), suggesting robustness of the classifier.



**Fig. 4.**

### DANCE attempted-copulation classifier outperforms existing methods of quantification.

(A) Attempted copulation index of males from 15 minute long videos from the ground-truth (grey), 'DANCE attempted copulation classifier' (orange) and MateBook (purple) against both mated and decapitated females. (B) Comparison of ground-truth vs. DANCE attempted-copulation classifier vs. MateBook (Kruskal-Wallis ANOVA with Dunn's test, ns,  $p > 0.9999$ ; \*\*\*\* $p < 0.0001$ ,  $n = 32$ ). (C) Regression analysis of the attempted-copulation classifier vs. ground-truth ( $R^2 = 0.8742$ ,  $n = 32$ ). (D) Regression analysis of MateBook vs. ground-truth ( $R^2 = 0.1512$ ,  $n = 32$ ). (E) F1 score, precision, and recall of DANCE and MateBook attempted-copulation classifiers against ground-truth scores.



**Fig. 5.**

### DANCE circling classifier outperforms existing methods of quantification.

(A) Circling index of males from 15 minutes long videos from the ground-truth (grey), 'DANCE circling classifier' (orange) and MateBook (purple) against decapitated virgin females. (B) Comparison of manual vs. DANCE vs. MateBook circling classifier (Ordinary one-way ANOVA with Dunnett's test, ns,  $p=0.8014$ ;  $*p=0.0157$ ,  $n=12$ ). (C) Regression analysis of the circling classifier vs. ground-truth ( $R^2=0.92$ ,  $n=12$ ). (D) MateBook vs. ground-truth ( $R^2=0.88$ ,  $n=12$ ). (E) F1 score, precision, and recall of DANCE and MateBook circling classifiers against ground-truth scores.

## Following

During ‘following’, the male follows around the female if it is moving in order to get closer to it and initiates the other courtship behaviors such as wing extension, licking, etc. (Spieth, 1968 [↗](#)). Since following is a relatively simpler behavior, both MateBook and the ‘DANCE following classifier’ performed well when compared to ground-truth (**Figure 5—figure supplement 2A-D** [↗](#)). However, the DANCE classifier had better precision, recall, and F1 scores (91.16%, 91.07%, and 91.12%) as compared to MateBook (65.77%, 83.39%, and 73.53%), indicating lower number of false positives and false negatives (**Figure 5—figure supplement 2E** [↗](#)).

## DANCE hardware

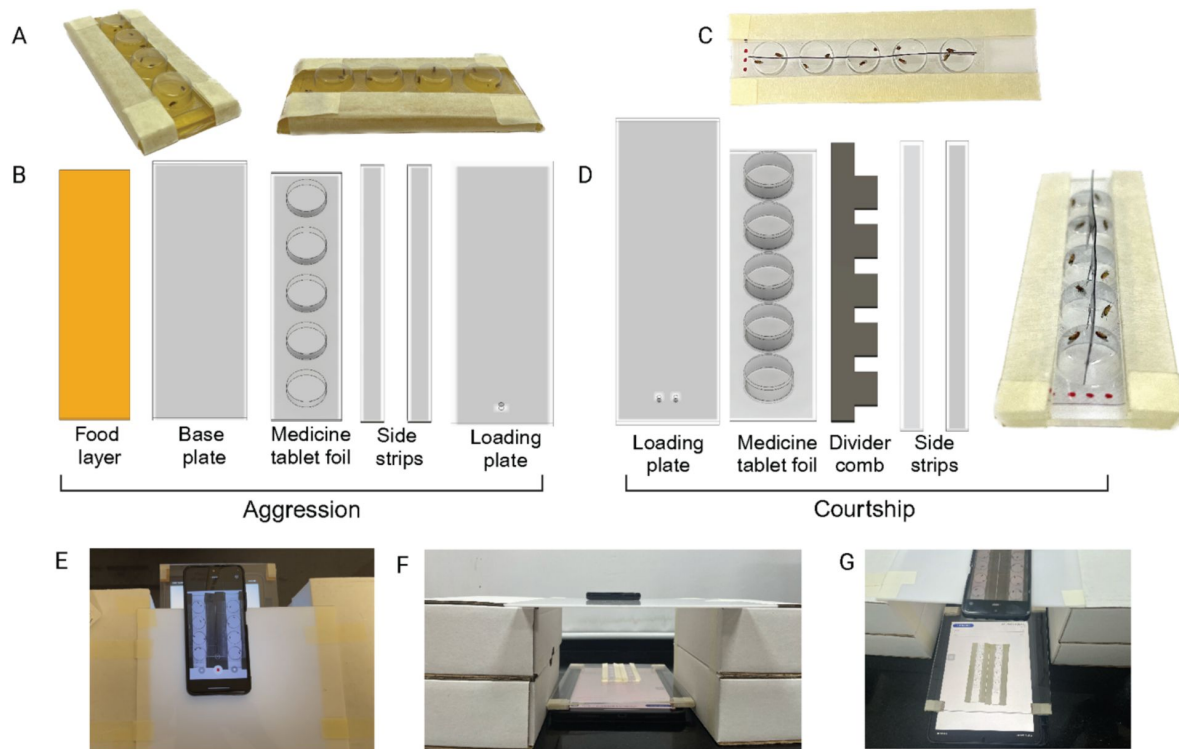
To measure aggression and courtship in flies, existing setups (**Figure 1A** [↗](#), **Figure 1—figure supplement 1** [↗](#)-2) (Dankert et al., 2009 [↗](#); Koemans et al., 2017 [↗](#)) have several bottlenecks, making it challenging to adopt these assays by the broader neuroscience community. For instance, it requires complex and specialized hardware (Dankert et al., 2009 [↗](#); Koemans et al., 2017 [↗](#)), necessitating access to sophisticated machine shops; or 3D printers to fabricate behavioral arenas (Chowdhury et al., 2021 [↗](#); Gil-Martí et al., 2023 [↗](#)). Setting up aggression assays (Dankert et al., 2009 [↗](#); Dierick, 2007 [↗](#)) requires coating chambers with fluon to prevent flies from walking on the walls, which is tedious. Recording the videos require expensive machine vision cameras and machine vision backlights and technical expertise for data acquisition and processing. To overcome these challenges, we devised the DANCE hardware (**Figure 6** [↗](#)), which provides an inexpensive, easy, scalable, and robust alternative to existing methods of recording *Drosophila* aggression and courtship behaviors.

The DANCE hardware consists of easy to find off-the-shelf components including transparent medicine tablet foils/blister packs of different diameters, which are repurposed to serve as arenas for recording behaviors; acrylic sheets; and paper tape (**Figure 6A-D** [↗](#), Supplementary video 1-2). To substitute expensive machine vision cameras, we used widely available Android smartphone cameras for recording and expensive machine vision backlights were replaced with electronic tablets or smartphones running white light application, to provide uniform backlight/illumination source (**Figure 6E-G** [↗](#)). Aggression and courtship behaviors were recorded at 30 fps with 1080 pixel resolution. In the DANCE aggression hardware (**Figure 6A-B, E** [↗](#)) medicinal foil is slid over a base plate containing an apple juice agar food layer over which flies fight (Supplementary Video 1-2). For DANCE courtship chambers (**Figure 6C-D, F-G** [↗](#)), the medicine tablet foil is slit from the middle using a razor, through which a thin ‘separator comb’ is inserted, which is made using x-ray film. This ‘separator comb’ keeps the males and females separate until the beginning of the recording, when it is manually removed (Supplementary video 3-4).

Since the heat generated from the tablet/smartphone screen during recording can affect the aggression and courtship behaviors, we placed a transparent acrylic sheet with an air- gap of 4 mm on top of the backlight screen to facilitate heat dissipation (**Figure 6E-G** [↗](#); Supplementary Video 2, 4), which proved to be crucial to obtain robust data (**Figure 7** [↗](#)).

## Benchmarking DANCE hardware

We used our DANCE classifiers to quantify aggression and courtship behaviors and found that wild type flies display these behaviors at similar levels in DANCE hardware as in the existing set ups (**Figure 7**, Supplementary video 5). We investigated the effects of social isolation vs. enrichment on aggression and courtship behaviors, since single-housing (SH) when compared to group housing (GH), increases courtship attempts (Dankert et al., 2009 [↗](#); Kim and Ehrman, 1998 [↗](#); Pan and Baker, 2014 [↗](#)), and promotes aggression (Agrawal et al., 2020 [↗](#); Wang et al., 2008 [↗](#); Yadav et

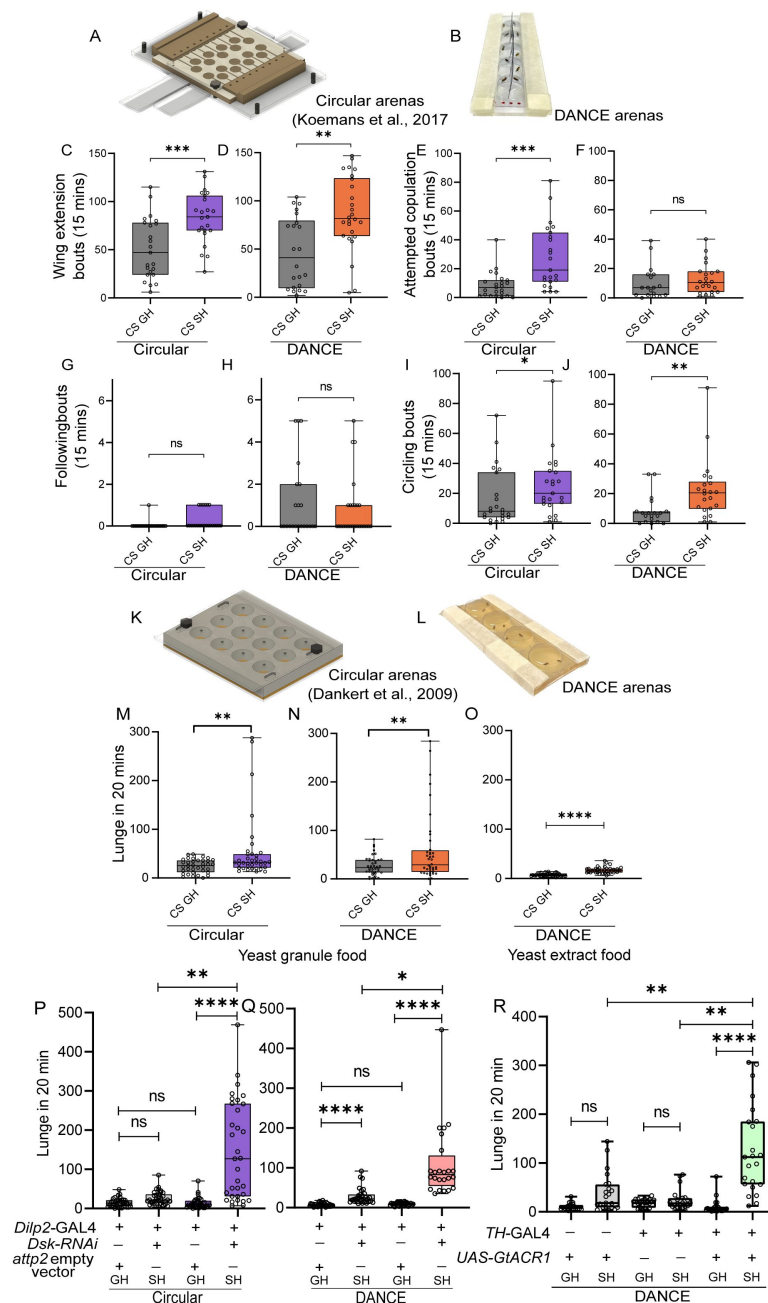


**Fig. 6.**

### **DANCE hardware and recordings setup.**

(A) DANCE aggression set up (B) 3D- rendered components of the aggression set up (C) DANCE courtship set up (D) 3D- rendered components of the courtship set up, male and female are separated on either side using X-ray film separator or 'divider comb'. (E-G) Top and side views of the DANCE setup with a smartphone camera for recording and electronic tablet being used as a backlight.





**Fig. 7.**

### Benchmarking DANCE hardware and testing various neurogenetic tools.

(A- B) Courtship behaviors recorded using a pre-existing (circular) and DANCE set up, from group-housed (GH) and single-housed (SH) flies, for (C-D) wing extension, (C) \*\*\* $p < 0.0010$ ,  $n = 23$ ; (D) \*\* $p < 0.0013$ , GH,  $n = 22$  and SH,  $n = 26$ . (E-F) Attempted copulation, (E) \*\*\* $p < 0.0002$ ,  $n = 23$ ; (F) ns,  $p < 0.1907$ , GH,  $n = 18$  and SH,  $n = 22$ . (G-H) Following, (G) ns,  $p > 0.0959$ ,  $n = 23$ ; (H) ns,  $p < 0.6589$ , GH,  $n = 22$  and SH,  $n = 26$ . (I-J) Circling, (I) \* $p < 0.012$ ,  $n = 23$ ; (J) \*\* $p < 0.0021$ , GH,  $n = 19$  and SH,  $n = 22$ . (K-L) Aggressive lunges recorded using a pre-existing (circular) and DANCE set up. (M-N) Lunges in SH flies compared to GH flies reared on food with yeast granules, (M) \*\* $p < 0.0138$ ,  $n = 36$ ; (N) \*\* $p < 0.0372$ ,  $n = 40$ . (O) Effect of yeast extract food on aggressive behavior; \*\*\*\* $p < 0.0001$ ,  $n = 38$ . (P-Q) Genetic knockdown of the neuropeptide Drosulfakinin (Dsk) in insulin-producing neurons using *dilp2-GAL4*. (P) ns,  $p < 0.0502$ , ns,  $p > 0.9999$ , \*\*\*\* $p < 0.0001$ , \*\* $p < 0.0040$ ,  $n = 35$ . (Q) \*\*\*\* $p < 0.0001$ , ns,  $p > 0.9999$ , \*\*\*\* $p < 0.0001$ , \* $p > 0.0210$ ,  $n = 30$ . (R) Optogenetic silencing of dopaminergic neurons by *UAS-GtACR1* driven by *TH-GAL4* driver, ns,  $p < 0.0986$ , ns,  $p > 0.9999$ , \*\*\*\* $p > 0.0001$ , \*\* $p > 0.0012$ , \*\* $p > 0.0013$ ,  $n = 24$ . (C-J and M-O) Mann-Whitney U test; (P-R) Kruskal-Wallis test with Dunn's multiple comparisons.

al., 2024 [↗](#)). We found comparable performance between pre-existing courtship setup (Koemans et al., 2017 [↗](#)) (**Figure 7A** [↗](#)) and the DANCE hardware (**Figure 7B** [↗](#)), both of which captured significant differences in multiple courtship behaviors in SH vs. GH males (**Figure 7C-J** [↗](#)).

Comparison of pre-existing set up for aggression (Dankert et al., 2009 [↗](#)) (**Figure 7K** [↗](#)) with the DANCE hardware (**Figure 7L** [↗](#)), showed that both the set ups faithfully captured even low levels of aggressive lunges in the wild type SH flies (**Figure 7M-N** [↗](#)). Availability of food affects *Drosophila* aggression (Lim et al., 2014 [↗](#)) and yeast in fly food was shown to interact with gut microbiome to affect male aggression (Jia et al., 2021 [↗](#)). We therefore investigated if the type of yeast used in food would affect aggression. Substituting yeast granules in fly food with yeast extract powder reduced the baseline aggression when flies were reared and differentially housed on food with yeast extract powder (**Figure 7O** [↗](#)). To investigate the effect of genetic knockdown using RNAi, which is a widely used tool in *Drosophila* genetics, we downregulated neuropeptide *Drosulfakinin* (Dsk) in insulin-producing neurons using *dilp2- GAL4*. Consistent with a previous finding (Agrawal et al., 2020 [↗](#)), aggressive lunges were significantly increased in SH males (**Figure 7P-Q** [↗](#)). Behavioral screens often utilize optogenetics to identify neuronal circuits responsible for aggression (Hoopfer et al., 2015 [↗](#); Wohl et al., 2023 [↗](#); Yadav et al., 2024 [↗](#)). It was earlier shown that while specific dopaminergic neurons may play a role in aggression, constitutive inactivation of a larger set of dopaminergic neurons using *TH-GAL4* led to unhealthy flies with defects in locomotion due to which they did not fight (Alekseyenko et al., 2013 [↗](#)). We reasoned that modern optogenetic tools may overcome the limitations of constitutive silencing and sought to investigate the role of dopaminergic neurons using our DANCE set up. We carried out optogenetic silencing during 20 minute interactions, by driving green light sensitive anion channelrhodopsin- *GtACR1* (Govorunova et al., 2015 [↗](#); Mohammad et al., 2017 [↗](#)) using *TH- GAL4*, and found significant increase in aggressive lunges in SH flies (**Figure 7R** [↗](#)). We also observed higher wing flicks as well as high intensity aggressive behaviors such as boxing and tussling (Supplementary Video 6). We did not find significant changes in locomotor activity between GH and SH flies despite 12 hours of continuous silencing (**Figure 7— figure supplement 1** [↗](#)). Together this suggests that the DANCE set up can be used with most of the common neurogenetic tools to screen for underlying molecular and neuronal circuitry governing aggression and courtship behaviors.

## Discussion

Here, we present the DANCE assay, an easy to use and robust analysis pipeline and inexpensive hardware to record and quantify aggression and courtship behaviors. We developed six novel behavioral classifiers using supervised learning to accurately quantify aggression and courtship behaviors of *Drosophila* males. We developed a versatile, easy to fabricate and inexpensive hardware setup costing <\$0.30, about 10,000 fold improvement over existing set ups, using easily available, off-the-shelf components. The performance of DANCE set up is comparable with more specialized and expensive existing setups. Therefore, DANCE enables rapid behavioral screening and wider adoption by the neuroscience community including in resource limited settings such as undergraduate education.

Various components of the DANCE assay including the behavioral classifiers, hardware design and analysis codes are publicly available and can be used independently as well. This offers flexibility to the neuroscience community to further customize the DANCE classifiers for specific needs or incorporate additional data without having to develop a classifier from scratch. Our study can also serve as a template for development of future behavioral classifiers for *Drosophila* aggression such as fencing, wing flick, box, tussle, chase, female headbutt etc. (Chen et al., 2002 [↗](#); Nilsen et al., 2004 [↗](#)) and courtship behaviors, such as- tapping by male, licking the posterior abdomen of the female, female rejection etc. (Sokolowski, 2001 [↗](#)).

Such high resolution analysis of complex social interactions could give us insights into mating dynamics, sexual selection and the influence of genetics and evolution. Further, the behavioral dynamics can help uncover sequencing of dynamic components of behaviors (Nilsen et al.,

2004 [↗](#); Seeds et al., 2014 [↗](#); Simon and Heberlein, 2020 [↗](#); Zhang et al., 2020 [↗](#)). It can also serve as a template for studying complex behaviors of various species of *Drosophila* and other insects. The ease of adaptability and portability of the DANCE assay can also be useful for ethologists studying insect behaviors closer to their natural habitat.

## Materials and Methods

The details of all the custom codes, analysis pipeline, sample files to run the analysis and DANCE classifiers are available in our GitHub repository at- <https://github.com/agrawallab/DANCE> [↗](#).

### Fly Husbandry

Flies were reared on standard food at 25°C and 65% relative humidity with a 12hr:12hr light-dark cycle. All assays were performed at 25°C with 65% relative humidity, unless mentioned otherwise. For aggression and courtship experiments, Canton-S male flies were collected within 24 hours of eclosion and housed in a group (20 male flies per vial, 90mm length and 25mm diameter) or isolated (1 male fly per vial, 70mm length and 10mm diameter) for 6 days.

The following fly lines were acquired from the Bloomington *Drosophila* Stock Center (BDSC), USA: *TH-GAL4*, *Dilp2-GAL4* (RRID:BDSC\_37516), *Dsk-RNAi* (RRID:BDSC\_25869) and attP2 empty vector control (RRID:BDSC\_36303). The *UAS-GtACR1* flies were a gift from Gaurav Das, NCCS, Pune, India and Canton-S (CS) flies were obtained from Ulrike Heberlein, HHMI, Janelia Research Campus, USA.

### Aggression assay

The traditional aggression assay was performed as described previously (Dankert et al., 2009 [↗](#); Dierick, 2007 [↗](#)). In brief, the behavioral chamber is made up of 12 well-aggressive arenas (10 mm in height and 16 mm in diameter/arena). These arenas were covered by a sliding lid with 2mm loading holes to facilitate the introduction of flies. A pair of male flies that were housed either as a group (GH) or single fly (SH) were introduced into the arena wells by gentle mouth aspiration through the loading holes. After the flies were loaded, the sliding lid was tightened using screws. Fluon (Insect-a-slip, Bioquip: cat #2871B) was coated on the arena walls and was allowed to dry overnight to create a slippery surface, thereby preventing the flies from climbing the walls. Sigmacote (Sigma-Aldrich: SL2) was used to coat the sliding lid of the arena to prevent flies from walking on the arena ceiling. The chamber was placed on the food plate containing commercial apple juice (without added sugars), 2.5% w/v of sucrose, and 2.25% w/v of agarose. For experiments to test effects of fly food nutrients, either 2.4% yeast extract powder (HiMedia: RM0271) or 2.4% Yeast granules (Prime Instant Dry yeast, AB Maury, India) were mixed in fly food. For optogenetic silencing experiments, 520-540 nm wavelength of LEDS (Lumileds:2835) were used at 0.0004  $\mu$ W intensity, measured using a Power meter (Newport:843-R). For all aggression assays, flies were allowed to acclimatize in the arena for 5 minutes, and then the activity was filmed for 20 minutes. The assays were performed during ZT0 - ZT2.5 i.e. first 2.5 hours of the morning activity peak.

### Courtship assay

Traditional courtship assays were performed as described by (Koemans et al., 2017 [↗](#); Ribeiro et al., 2018 [↗](#)). We introduced single pairs of male and female flies using an aspirator in an 18-well courtship chamber, consisting of individual wells/arenas with a diameter of 10 mm. Male and female flies were introduced into one half of the chamber using the sliding entry holes with a removable separator that divided the chamber into two halves. The flies were allowed to acclimatize to the arena for 5 minutes, after which the separator was removed, and the courtship behavior was recorded for 15 minutes at 30 fps using a white backlight. The assay was performed

from ZT0 to ZT3 or ZT9 to ZT12 (during peak activity windows). For generating mated females, we housed 20 females with 10 males together in a single vial for 4-6 days. For decapitated virgins, 2-4 days old virgin females were collected in a vial, and the decapitation was performed after anesthetizing on a CO<sub>2</sub> fly-pad, just before the assay. The assays were performed during ZT0-ZT2.5 i.e., first 2.5 hours/morning activity peak or ZT9-ZT11.5 i.e. last 2.5 hours/evening activity peak.

## MateBook

To quantify courtship data, an automated software called MateBook was used (<https://github.com/Dicksonlab/MateBook>) (Ribeiro et al., 2018). MateBook tracks the fly's location and position for each video frame using the machine-vision approach. The software has built-in classifiers for each of the behaviors of a male fly during courtship, such as following, wing extension, orientation, copulation, and circling. The quantified data is generated in the form of a .tsv file, which contains bouts and other parameters related to courtship behavior. The software also provides an ethogram, which is a color-coded timeline of all the behaviors.

## DANCE hardware

Circular, transparent medicinal tablet foils/blister packs were used to serve as aggression or courtship arenas, and acrylic sheets were used as base plates, and the DANCE hardware setup is held together using paper tape. The DANCE aggression arena was 13 mm in diameter and 4 mm in height. A thin acrylic base plate (2 mm thickness) with a loading hole (2 mm diameter) and a base plate with food and a side spacer were held together using paper tape (Supplementary Video 1). The tip of the food plate was also covered using paper tape for easy sliding of the medicinal tablet foil without damaging the food layer.

Sigmacote (Sigma-Aldrich: SL2) was used to coat the walls and roof of the medicine blister foil using cotton swab (Solimo, Amazon India), to prevent flies from climbing the walls of the arena. A pair of male flies were introduced into the medicinal foil wells by gentle aspiration through the loading holes. After loading flies into the medicinal foil wells, the foil was slid onto the food bed with the help of side spacers and tape on the tip of the food bed. After sliding it, the assembly was taped from ends to avoid gaps and prevent flies from escaping the arenas (Supplementary Video 2).

The DANCE courtship arena used transparent medicine blister packs with a diameter of 11mm and a height of 4.5 mm. The top surface across all arenas (5 in a row) had thin slits cut using a sharp razor to fit in the separator comb which was built by cutting used x-ray film. The foil and base plate (2 mm thickness) with loading hole (2 mm diameter) was held together by a paper tape. The base plate for courtship had two holes spaced apart in such a way that, male and female flies can be loaded on either side of the separator comb. The assembly was taped from ends to avoid gaps and prevent flies from escaping the arenas.

Once the setup was ready to record, the separator comb was gently lifted up, while ensuring the slit edges stay intact to avoid distortions during recordings (Supplementary videos 3-4).

## Video acquisition

For the traditional set ups, the interaction of the flies was recorded using machine-vision cameras (DMK 33UX252 USB 3.0 monochrome camera). A white backlight (TMS, BHS4- 00-100-X-W-24V) provided the light source for both the courtship and aggression experiments. Videos were recorded at 30 frames per second (fps) for 15 minutes for courtship or 20 minutes for aggression in H.264 (.mp4) format with 1440x1080 resolution. These videos were used for training, testing, and validating the DANCE classifiers.

Alternatively, for DANCE hardware testing, we used various Android smartphone cameras (Huawei Y9 2019; OnePlus Nord CE 2 Lite 5G, model: CPH2381 or Redmi Note 11 Pro+ 5G, model: 221116SI) at 30 fps with 1080p resolution in H.264 (.mp4) format for 15 minutes (courtship) or 20 minutes (aggression) with electronic tablet (iPad Air, 5<sup>th</sup> Generation) or smartphone (iPhone 13) running a ‘white screen light app’ as the background illumination source. A transparent acrylic sheet with 4 mm spacers was kept on top of this ‘backlight’ to create an air gap to ensure heat exchange and to prevent the DANCE arenas from getting hot.

## Tracking flies using FlyTracker

To successfully classify the behavior of the flies, we tracked their location and body parts using Caltech FlyTracker (Eyjolfsson et al., 2014). It allows for detection and tracking of flies based on their location, body position, orientation, and interactions across the frames. Occurrence of identity switches between male and female flies was corrected using the ‘identity correction’ tool available in the “visualizer” program of the FlyTracker package. This data was then pushed to the JAABA pipeline to develop DANCE classifiers.

## Developing the DANCE behavioral classifiers

The Janelia Automatic Animal Behavior Annotator (JAABA) was used to develop DANCE behavioral classifiers. The algorithm allows experienced users to ‘encode their intuition’ to develop ‘classifiers’ that help to annotate various animal behaviors using tracking data from the behavioral recordings (Kabra et al., 2013).

Tracking data generated using FlyTracker from training videos were processed using JAABA. The stereotypic behavior bouts were labeled as ‘true behavior’, e.g., ‘Lunge’, ‘Wing extension’ etc., and an approximately equal number of obvious non-behavior bouts (‘None’), were labeled. After each round of training, the false positives predicted by the classifier would be specifically labeled as ‘None.’ This cycle of training and correcting the predictions was done until there was no improvement in the classifier’s performance. We included courtship videos of males with decapitated virgins in the training set for the attempted copulation and circling classifiers to increase the number of labeled bouts of True behavior, as these behaviors occur more frequently with decapitated virgins. Additionally, we included videos with decapitated virgins in the training sets for the following and copulation classifiers to provide more examples of the ‘None’ behavior. For the wing extension classifier, we used only mated females, as there were an adequate number of bouts representing both the true behavior and the ‘None’ behavior.

Various DANCE classifiers were developed using following frames: lunge: 2730 frames (91 seconds in total); wing extension: 99,947 frames (3,331.57 seconds in total); attempted- copulation: 39,513 frames (1,317.1 seconds in total); copulation: 56,979 frames (1899.3 seconds in total); circling: 14,396 frames (479.87 seconds in total); following: 25,787 frames (859.57 seconds in total).

DANCE classifiers, custom codes and other analysis details are available in our GitHub repository at- <https://github.com/agrawallab/DANCE>.

## Characterization of male aggression and courtship behaviors

**Lunge:** It is defined as ‘the attacking fly rises on his hind legs, lifting his long body axis by 45° then snaps down on his opponent with his head reaching a velocity of 200 mm/s’ (Dankert et al., 2009).

Courtship classifiers were developed for five male courtship behaviors based on existing definitions (Dankert et al., 2009; Ribeiro et al., 2018). For the duration-based classifiers, a filter of minimum bout length was added in the post-processing of each classifier as a measure to



eliminate the bouts shorter than 98% of manual bouts. Details of individual courtship classifiers are mentioned below.

**Wing Extension:** Dankert et.al 2009, Ribeiro et al., 2018 [\[1\]](#) define a wing extension, when the angle between the body's main axis and the line from the body's center to the tip of the wing exceeds 30°, with the behavior persisting for approximately 13 frames i.e. 0.5s, at 25 fps. We processed our data using MateBook as described (Ribeiro et al., 2018 [\[1\]](#)); but with a reduced persistence filter for wing extension to 0.33 seconds from the default 0.50 seconds to match our classifier's threshold of minimum 10 frames, since in all the courtship videos that we analyzed, >98% of the bouts followed this threshold.

**Attempted copulation and copulation:** Existing definitions, such as (Ribeiro et al., 2018 [\[1\]](#)) define copulation as 'occlusions' persisting for more than 45 seconds and does not quantify 'attempted-copulation'. We used a similar threshold while training our classifiers but ensured that it is not merely an 'occlusion'— if the male curls its abdomen without mounting or mounts the female for a minimum of 0.33 seconds (10 frames at 30 fps) to a maximum of <45 seconds, it was considered as 'attempted-copulation' whereas mounting for >45 seconds (>1350 frames at 30 fps) was considered as 'copulation'. The results were compared with MateBook by matching its 'event settings'.

**Circling:** (Dankert et al., 2009 [\[1\]](#)) define circling as the fly drifting sideways in a circle with approximately constant velocity. Additionally, (Ribeiro et al., 2018 [\[1\]](#)) considered this behavior with a persistence of 13 frames (0.5s at 25 fps). We developed our 'DANCE circling classifier' based on these existing definitions, but with a reduced persistence of 10 frames (0.33s at 30 fps) and compared the results with MateBook by matching its 'event settings'.

**Following:** Ribeiro et al., 2018 [\[1\]](#) define the event following as when the 'the male keeps distance to the female between 2 and 5mm, while directly behind the female, while both flies are walking with a minimum speed of 2mm/s' with the behavior persisting for 25 frames i.e. 1.0s, at 25 fps. We developed our 'DANCE following extension classifier' based on these existing definitions, but with a reduced persistence of 10 frames (0.33s, at 30 fps) and compared the results with MateBook by matching its 'event settings'.

## Manual behavioral annotations

To quantitatively measure the classifier's performance, manual behavioral annotations, or 'ground-truthing' were performed using the JAABA's ground-truthing mode (Kabra et al., 2013 [\[1\]](#)). The classifier's performance is measured by its robustness in predicting the behavioral bouts for the frames it was not trained on, which comprises the 'testing set.' In the testing set, we first annotate the behavioral bouts as 'true behavior' without looking at the predictions of the output. Then these videos are processed with the same classifier using JAABAPlot and compared as described subsequently.

## Comparison of manual and DANCE annotations

The classifier's output was compared with that of the manual ground-truth, based on both the number of bouts for the one-frame lunge classifier and the bout duration for all duration-based courtship classifiers (wing extension, following, circling, attempted-copulation, and copulation). The duration of the bouts was used to derive the respective behavioral index. A regression analysis and classification metrics were calculated between the bout numbers or the behavior index based on the type of classifier for each of the ground-truth, DANCE, and the existing software's scores using Prism 8 (GraphPad Software).

The manual scores and JAABA output were compared, bout by bout, for the total number of true positives, false positives, and false negatives. Human annotated bouts (ground-truth) were used as a reference to the number of true positives. A ‘true positive’ is called when a JAABA bout overlaps with ground-truth bout for one or more frames (over 0.03 ms). If one ground-truth bout overlapped with two or more JAABA bouts, then all the JAABA bouts would be collectively counted as one bout, i.e., one true positive bout. Similarly, if two or more ground-truth bouts overlapped with a single JAABA bout, then the number of ground-truth bouts would be the count for the JAABA bout; i.e., two or more number of true positives.

When a JAABA bout does not overlap with any ground-truth (human annotation), then it would be considered a false positive. When a ground-truth bout does not overlap with any JAABA bout, it is considered a false negative (Leng et al., 2020 [DOI](#)).

We used a custom Python code, to calculate the number of total overlaps between the ground-truth, the DANCE classifier’s, and the existing algorithm’s output. Once we had the total number of true positives, false positives, and false negatives, we derived the precision, recall and F1 scores to gain a deeper understanding of the performance of DANCE classifiers. Precision rate represents the ratio of correctly predicted positive observations to the total predicted positives; the recall rate represents the ratio of correctly predicted positive observations to all actual positives; and the F1 score is a single metric which is a harmonic mean of both precision and recall. We used the following formulas to calculate precision, recall, and F1 score:

$$\begin{aligned} \text{precision} &= \frac{(\text{number of true positives})}{(\text{number of true positives}) + (\text{number of false positives})} \\ \text{recall} &= \frac{(\text{number of true positives})}{(\text{number of true positives}) + (\text{number of false negatives})} \\ \text{F1 score} &= 2 \times \frac{(\text{precision} \times \text{recall})}{(\text{precision} + \text{recall})} \end{aligned}$$

## Statistical analysis

For non-normally distributed data, Mann-Whitney U test or ANOVA with appropriate posthoc correction was used. Statistical analysis of behavioral data was performed using Prism 8 (GraphPad Software), custom scripts or Microsoft Excel.

## Acknowledgements

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## Additional information

Figure 1—Figure Supplement 1.

**Aggression chamber described by Dankert *et al.*, 2009**

It consists of a bottom food plate, 12 wells (aggression arenas), a top plate with fly loading holes and a screw slot for sliding the loading plate.

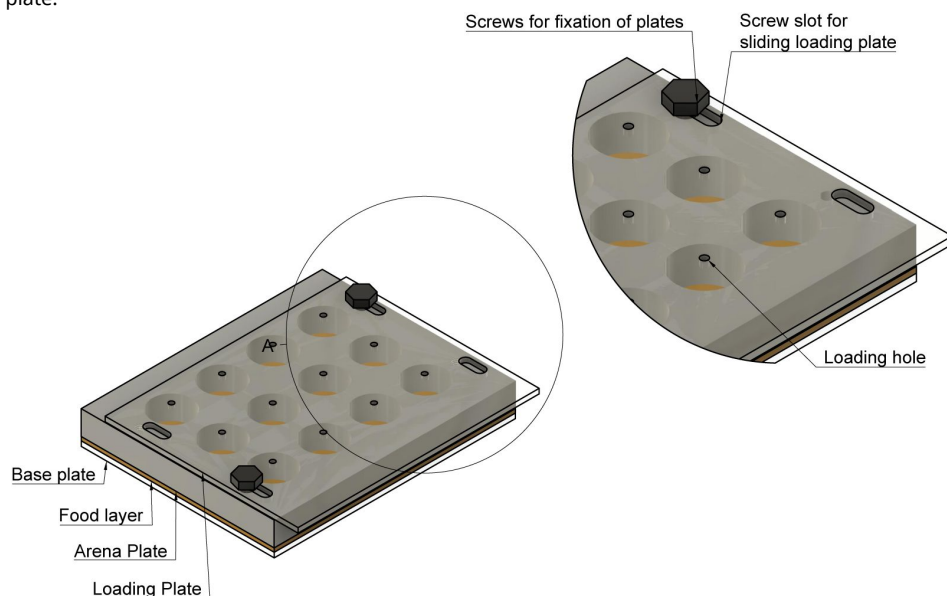
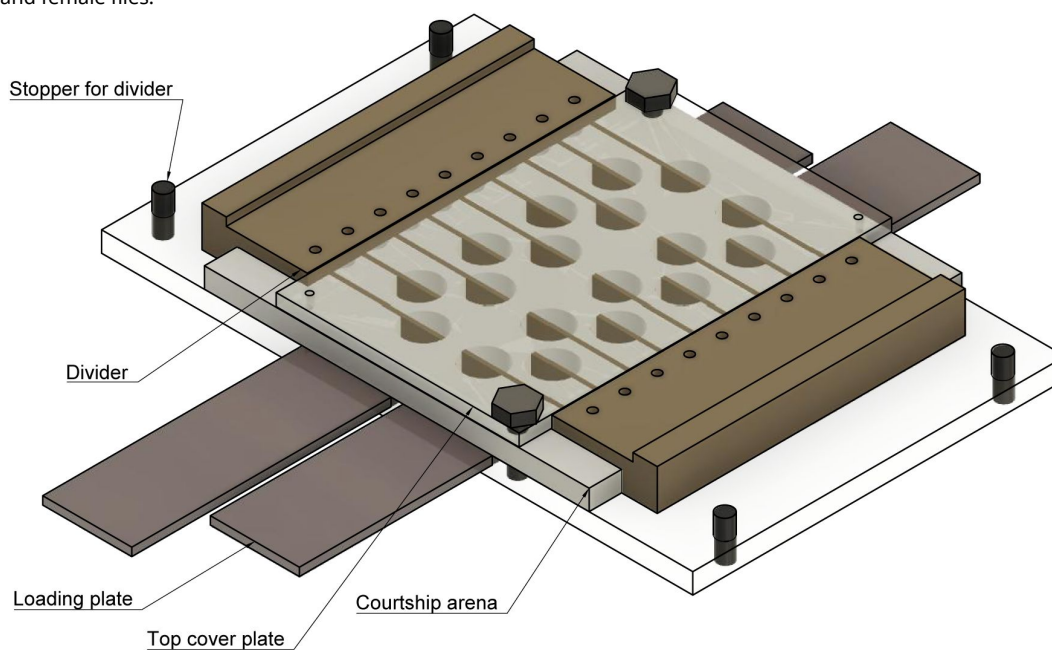
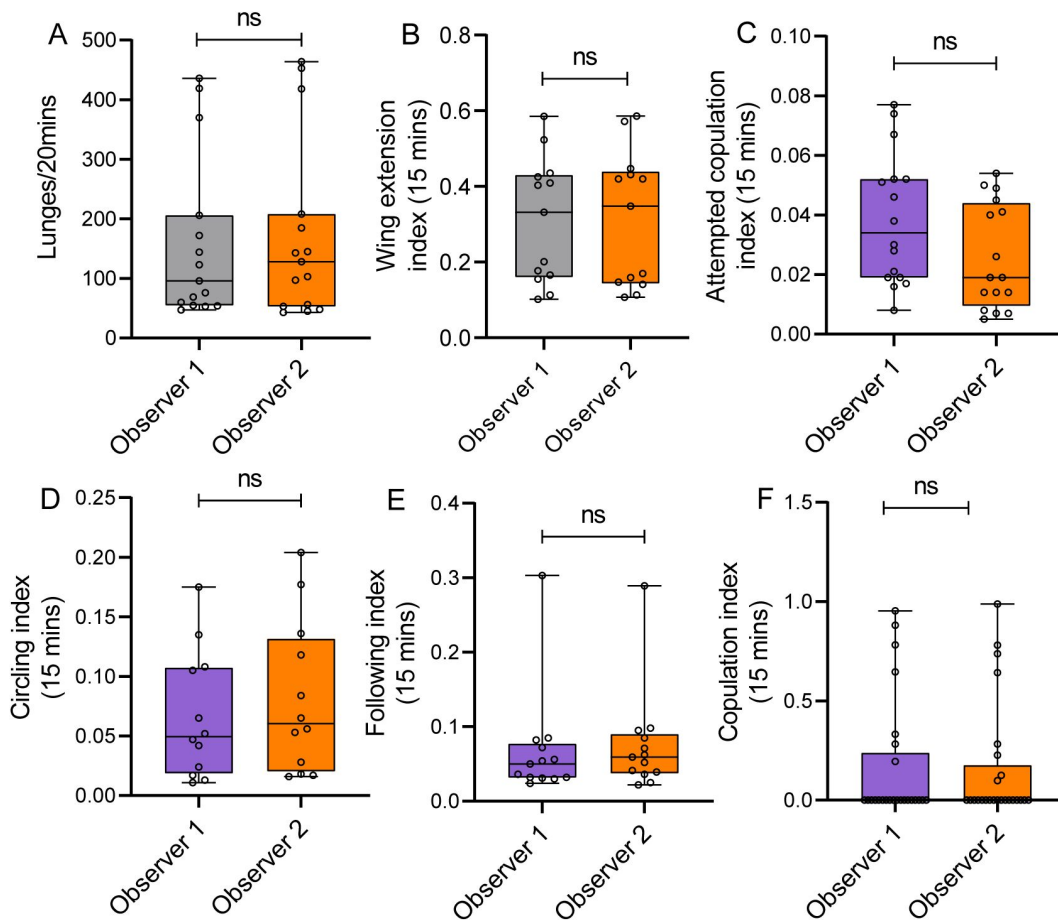


Figure 1—Figure Supplement 2.

**Courtship set-up described by Koemans *et al.*, 2017**

It consists of 18 wells (courtship arenas), a top cover plate, a sliding loading plate, and a sliding divider assembly to separate male and female flies.

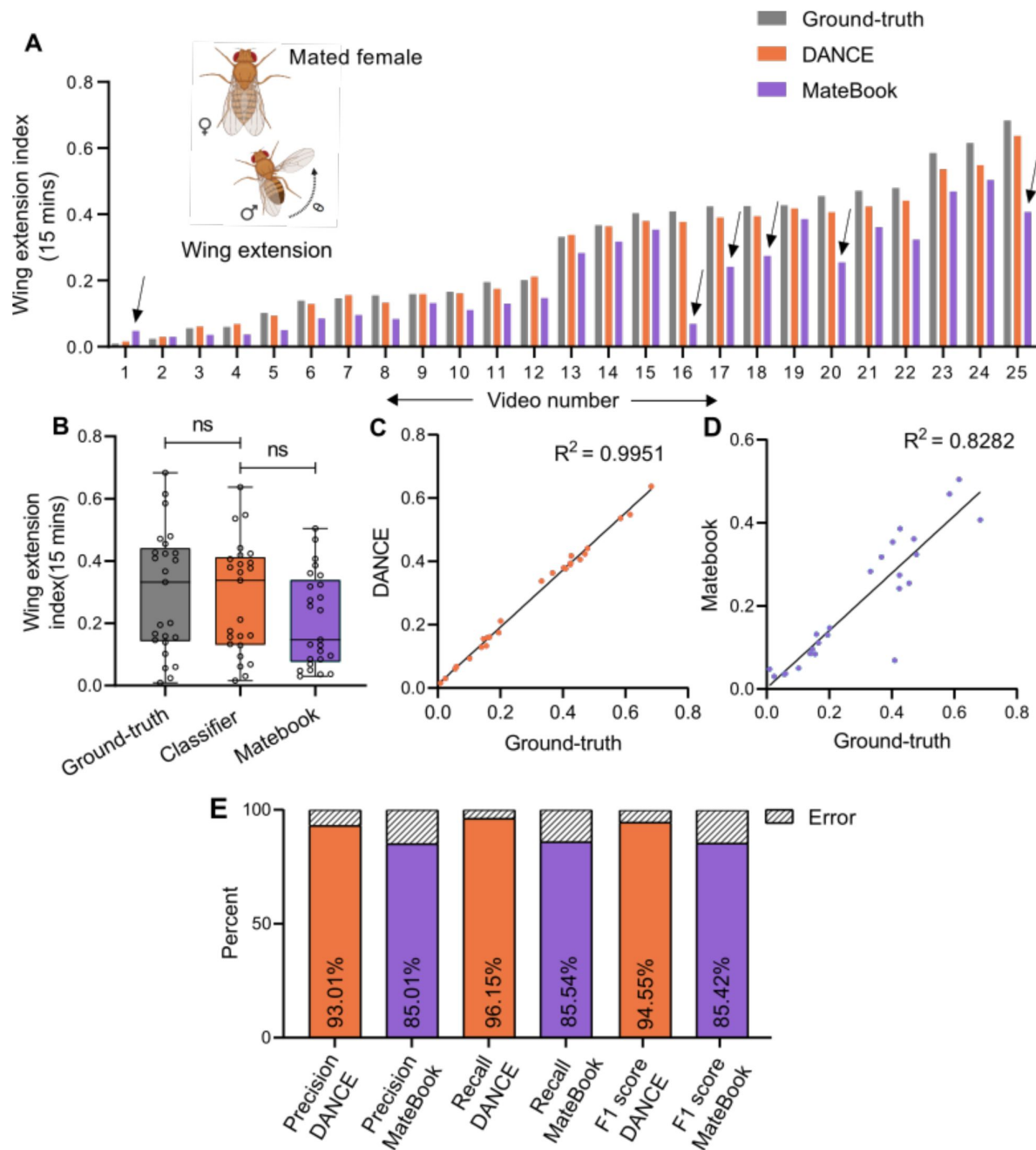




**Figure 2—Figure Supplement 1.**

**Comparison of annotations by two independent evaluators to assess observer bias during classifier ground-truthing.**

(A) Aggressive lunge,  $p=0.8789$ ,  $n=15$ . (B) Courtship including wing extension,  $p=0.9999$ ,  $n=13$ . (C) Attempted copulation,  $p=0.0571$ ,  $n=16$ . (D) Circling,  $p=0.4343$ ,  $n=12$ . (E) Following,  $p=0.4405$ ,  $n=13$ . (F) Copulation,  $p=0.9221$ ,  $n=25$ . (A-F) Mann-Whitney U test.

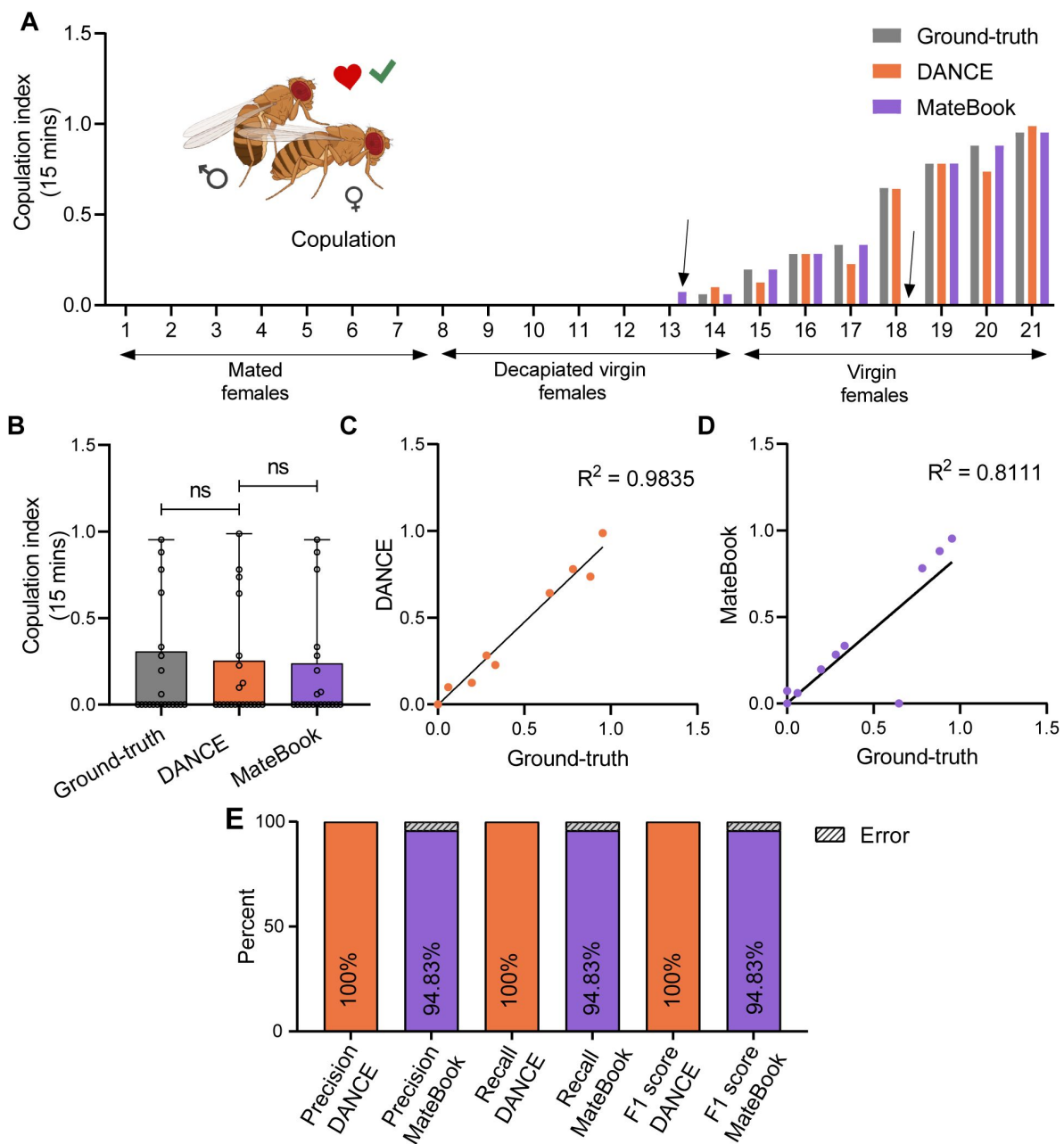


**Figure 3—Figure Supplement 1.**

### DANCE wing extension classifier outperforms existing quantification methods in videos with mated females.

(A) Wing extension of males from 15-minute-long videos from the ground-truth (grey), the DANCE classifier (orange) and MateBook (purple) against mated females. MateBook underscored wing extension across multiple videos (black arrows) (Friedman's ANOVA with Dunn's test,  $p=0.3582$ ;  $p<0.0001$ ,  $n=25$ ). (B) Comparison of ground-truth vs. DANCE vs. MateBook wing extension classifier (Kruskal-Wallis ANOVA with Dunn's test,  $p>0.9999$ ;  $p=0.1039$ ,  $n=25$ ). (C) Regression analysis of the DANCE classifier vs. ground-truth ( $R^2=0.9951$ ,  $n=25$ ). (D) MateBook vs. ground-truth ( $R^2=0.8282$ ,  $n=25$ ). (E) F1 score, precision, and recall of DANCE classifier and MateBook against ground-truth scores.

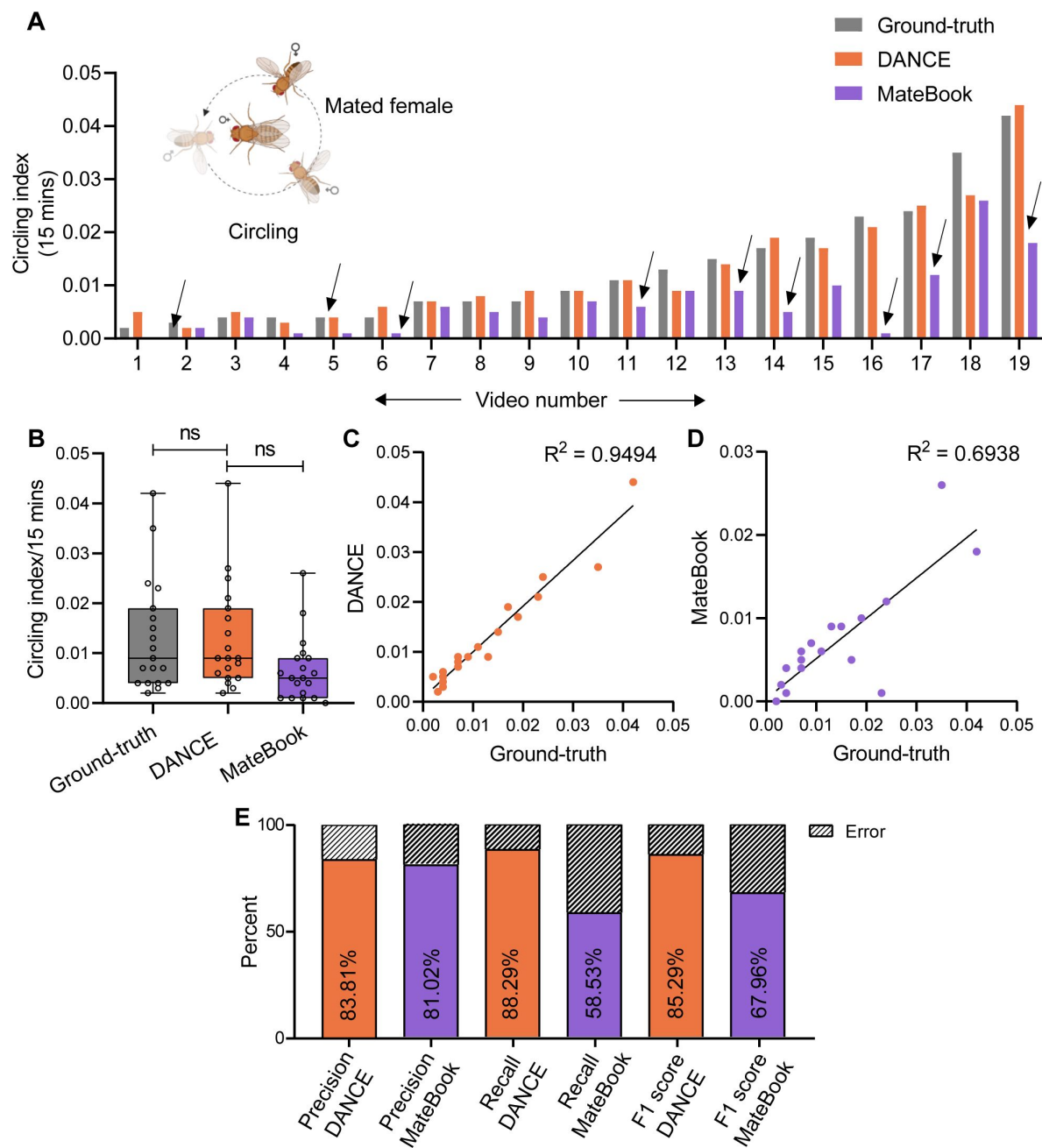




**Figure 4—Figure Supplement 1.**

#### DANCE copulation classifier evaluation in mixed female dataset.

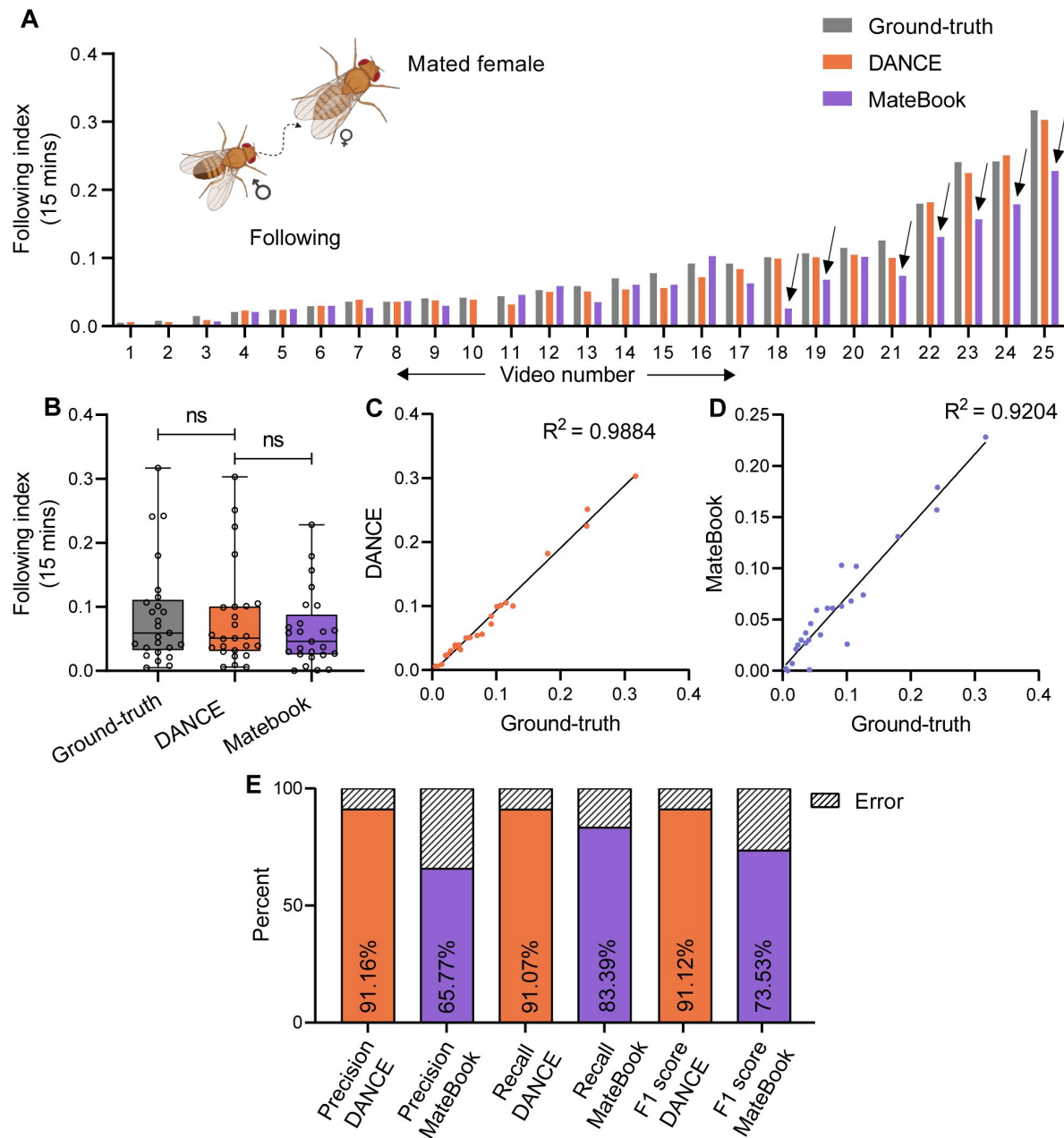
(A) Quantification of copulation in 15 minutes from individual videos showing scores from the manual method (grey), DANCE copulation classifier (orange), and MateBook (purple) (Friedman ANOVA with Dunn's test,  $p > 0.9999$ ;  $p > 0.9999$ ,  $n = 21$ ). (B) Box plot comparison of manual vs. DANCE copulation classifier vs. MateBook (Kruskal-Wallis ANOVA with Dunn's test,  $p > 0.9999$ ;  $p > 0.9999$ ,  $n = 21$ ). (C) Bar plots showing F1 score, precision, and recall of DANCE copulation classifier and MateBook against ground-truth scores. (D) Regression analysis of the copulation classifier vs. manual scores ( $R^2 = 0.98$ ,  $n = 21$ ). (E) MateBook vs. manual scores ( $R^2 = 0.81$ ,  $n = 21$ ).



**Figure 5—Figure Supplement 1.**

#### DANCE circling classifier evaluation in the mated female dataset.

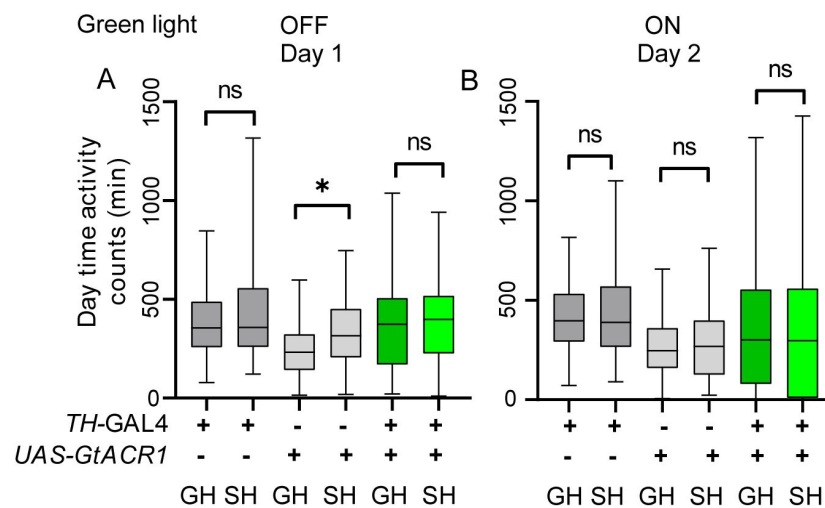
(A) Circling index of males from 15-minute videos from the ground-truth (grey), 'DANCE circling classifier' (orange) and MateBook (purple) against mated female dataset (Friedman's ANOVA with Dunn's test,  $p > 0.9999$ ;  $p < 0.0001$ ,  $n = 19$ ). (B) Comparison of manual vs. DANCE vs. MateBook circling classifier (Kruskal-Wallis ANOVA,  $p > 0.9999$ ;  $p = 0.0822$ ,  $n = 19$ ). (C) Regression analysis of the circling DANCE classifier vs. ground-truth ( $R^2 = 0.9494$ ,  $n = 19$ ). (D) MateBook vs. ground-truth ( $R^2 = 0.6938$ ,  $n = 19$ ). (E) F1 score, precision, and recall of DANCE and MateBook circling classifiers against ground-truth scores.



**Figure 5—Figure Supplement 2.**

#### DANCE following classifier evaluation in the mated female dataset.

(A) Following index of males from 15 minutes long videos from the ground- truth (grey), 'DANCE following classifier' (orange) and MateBook (purple) against mated females (Friedman's ANOVA with Dunn's test,  $p=0.1794$ ;  $p=0.0029$ ,  $n=25$ ). (B) Box plot comparison of manual vs. DANCE following classifier vs. MateBook (Kruskal-Wallis ANOVA with Dunn's test,  $p>0.9999$ ;  $p=0.5287$ ,  $n=25$ ). (C) Regression analysis of the following classifier vs. ground-truth ( $R^2=0.9894$ ,  $n=25$ ). (D) MateBook vs. ground-truth ( $R^2=0.9204$ ,  $n=25$ ). (E) F1 score, precision, and recall of DANCE and MateBook following classifiers against ground-truth scores.



**Figure 7—Figure Supplement 1.**

### Effect of optogenetic silencing of dopaminergic neurons on *Drosophila* activity.

(A, B) Transient silencing of dopaminergic neurons using *UAS-GtACR1* driver didn't affect daytime activity between SH and GH flies. (A) Without silencing mediated by green light on day 1; *TH-GAL4*, ns,  $p=0.7383$ , GH:  $n=43$ , SH:  $n=43$ ; *UAS-GtACR1*, ns,  $p=0.4812$ , GH:  $n=51$ , SH:  $n=42$ ; *TH-GAL4>UAS-GtACR1*, ns,  $p=0.9942$ , GH:  $n=54$ , SH:  $n=51$ . (B) With silencing mediated by green light on day 2; *TH-GAL4*, ns,  $p=0.9976$ , GH:  $n=43$ , SH:  $n=43$ ; *UAS-GtACR1*, ns,  $p=0.9779$ , GH:  $n=51$ , SH:  $n=42$ ; *TH-GAL4>UAS-GtACR1*, ns,  $p=0.9974$ , GH:  $n=54$ , SH:  $n=51$ . One-way ANOVA with Tukey's multiple comparisons test for within each day comparison. Two-way ANOVA for comparison across days; interaction, ns,  $p=0.5504$ , silencing, ns,  $p=0.5172$ , housing, ns,  $p=0.1602$ , GH:  $n=43$ , SH:  $n=41$ ; *UAS-GtACR1*, interaction, ns,  $p=0.4533$ , silencing, ns,  $p=0.3602$ , housing,  $*p=0.255$ , GH:  $n=51$ , SH:  $n=42$ ; *TH-GAL4>UAS-GtACR1*, interaction, ns,  $p=0.9977$ , silencing, ns,  $p=0.2454$ , housing, ns,  $p=0.5868$ , GH:  $n=54$ , SH:  $n=51$ .

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## Additional files

**Supplementary File 1.** Bill of material for DANCE set up and comparison with existing setups. [↗](#)

**Supplementary File 2.** Comparison between DANCE lunge classifier, ground-truth and existing methods. [↗](#)

**Supplementary File 3.** Definitions for behavioral classifiers. [↗](#)

**Supplementary File 4.** Comparison between DANCE courtship classifiers, ground-truth and MateBook. [↗](#)

**Supplementary Video 1.** 3D rendered DANCE Aggression hardware. [↗](#)

**Supplementary Video 2.** Using DANCE hardware setup for recording aggression. [↗](#)

**Supplementary Video 3.** 3D rendered DANCE Courtship hardware. [↗](#)

**Supplementary Video 4.** Using DANCE hardware setup for recording Courtship. [↗](#)

**Supplementary Video 5.** Aggression and courtship behaviors recorded in DANCE hardware. [↗](#)

**Supplementary Video 6.** Optogenetic silencing of dopaminergic neurons in DANCE shows increased aggression. [↗](#)

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**Reviewer #1 (Public review):**

The study introduces an open-source, cost-effective method for automating the quantification of male social behaviors in *Drosophila melanogaster*. It combines machine-learning-based

behavioral classifiers developed using JAABA (Janelia Automatic Animal Behavior Annotator) with inexpensive hardware constructed from off-the-shelf components. This approach addresses the limitations of existing methods, which often require expensive hardware and specialized setups. The authors demonstrate that their new "DANCE" classifiers accurately identify aggression (lunges) and courtship behaviors (wing extension, following, circling, attempted copulation, and copulation), closely matching manually annotated ground-truth data. Furthermore, DANCE classifiers outperform existing rule-based methods in accuracy. Finally, the study shows that DANCE classifiers perform as well when used with low-cost experimental hardware as with standard experimental setups across multiple paradigms, including RNAi knockdown of the neuropeptide Dsk and optogenetic silencing of dopaminergic neurons.

The authors make creative use of existing resources and technology to develop an inexpensive, flexible, and robust experimental tool for the quantitative analysis of *Drosophila* behavior. A key strength of this work is the thorough benchmarking of both the behavioral classifiers and the experimental hardware against existing methods. In particular, the direct comparison of their low-cost experimental system with established systems across different experimental paradigms is compelling. While JAABA-based classifiers have been previously used to analyze aggression and courtship (Tao et al., *J. Neurosci.*, 2024; Sten et al., *Cell*, 2023; Chiu et al., *Cell*, 2021; Isshi et al., *eLife*, 2020; Duistermars et al., *Neuron*, 2018), the demonstration that they work as well without expensive experimental hardware opens the door to more low-cost systems for quantitative behavior analysis.

Although the study provides a detailed evaluation of DANCE classifier performance, its conclusions would be strengthened by a more comprehensive analysis. The authors assess classifier accuracy using a bout-level comparison rather than a frame-level analysis, as employed in previous studies (Kabra et al., *Nat Methods*, 2013). They define a true positive as any instance where a DANCE-detected bout overlaps with a manually annotated ground-truth bout by at least one frame. This criterion may inflate true positive rates and underestimate false positives, particularly for longer-duration courtship behaviors. For example, a 15-second DANCE-classified wing extension bout that overlaps with ground truth for only one frame would still be considered a true positive. A frame-level analysis performance would help address this possibility.

In summary, this work provides a practical and accessible approach to quantifying *Drosophila* behavior, reducing the economic barriers to the study of the neural and molecular mechanisms underlying social behavior.

<https://doi.org/10.7554/eLife.105465.1.sa2>

#### **Reviewer #2 (Public review):**

##### **Summary:**

This manuscript addresses the development of a low-cost behavioural setup and standardised open-source high-performing classifiers for aggression and courtship behaviour. It does so by using readily available laboratory equipment and previously developed software packages. By comparing the performance of the setup and the classifiers to previously developed ones, this study shows the classifier's overperformance and the reliability of the low-cost setup in recapitulating previously described effects of different manipulations on aggression and courtship.

##### **Strengths:**



The newly developed classifiers for lunges, wing extension, attempted copulation, copulation, following, and circling, perform better than available previously developed ones. The behavioural setup developed is low cost and reliably allows analysis of both aggression and courtship behaviour, validated through social experience manipulation (social isolation), gene knock (Dsk in Dilp2 neurons) and neuronal inactivation (dopaminergic neurons) known to affect courtship and aggression.

#### Weaknesses:

Aggression encompasses multiple defined behaviours, yet only lunges were analysed. Moreover, the CADABRA software to which DANCE was compared analyses further aggression behaviours, making their comparisons incomplete. In addition, though DANCE performs better than CADABRA and Divider in classifying lunges in the behavioural setup tested, it did not yield very high recall and F1 scores.

DANCE is of limited use for neuronal circuit-level enquiries, since mechanisms for intensity and temporally controlled optogenetic manipulations, which are nowadays possible with open-source software and low-cost hardware, were not embedded in its development.

<https://doi.org/10.7554/eLife.105465.1.sa1>

#### Reviewer #3 (Public review):

The preprint by Yadav et al. describes a new setup to quantify a number of aggression and mating behaviors in *Drosophila melanogaster*. The investigation of these behaviors requires the analysis of a large number of videos to identify each kind of behavior displayed by a fly. Several approaches to automatize this process have been published before, but each of them has its limitations. The authors set out to develop a new setup that includes very low-cost, easy-to-acquire hardware and open-source machine-learning classifiers to identify and quantify the behavior.

#### Strengths:

(1) The study demonstrates that their cheap, simple, and easy-to-obtain hardware works just as well as custom-made, specialized hardware for analyzing aggression and mating behavior. This enables the setup to be used in a wide range of settings, from research with limited resources to classroom teaching.

(2) The authors used previously published software to train new classifiers for detecting a range of behaviors related to aggression and mating and to make them freely available. The classifiers are very positively benchmarked against a manually acquired ground truth as well as existing algorithms.

(3) The study demonstrates the applicability of the setup (hardware and classifiers) to common methods in the field by confirming a number of expected phenotypes with their setup.

#### Weaknesses:

(1) When measuring the performance of the duration-based classifiers, the authors count any bout of behavior as true positive if it overlaps with a ground-truth positive for only 1 frame - despite the minimal duration of a bout is 10 frames, and most bouts are much longer. That way, true positives could contain cases that are almost totally wrong as long there was an overlap of a single frame. For the mating behaviors that are classified in ongoing bouts, I think performance should be evaluated based on the % of correctly classified frames, not bouts.

(2) In the methods part, only one of the pre-existing algorithms (MateBook), is described. Given that the comparison with those algorithms is a so central part of the manuscript, each of them should be briefly explained and the settings used in this study should be described.

Taken together, this work can greatly facilitate research on aggression and mating in *Drosophila*. The combination of low-cost, off-the-shelf hardware and open-source, robust software enables researchers with very little funding or technical expertise to contribute to the scientific process and also allows large-scale experiments, for example in classroom teaching with many students, or for systematic screenings.

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